DEGREE PROJECT

A Lot with A Little

Evaluating Usability Methods for Developing Two New Methods for Small Test Groups

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Industrial Design Engineering, master's level
2026

Luleå University of Technology
Department of Social Sciences, Technology and Arts

LULEÅ UNIVERSITY OF TECHNOLOGY

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CIVILINGENJÖR I TEKNISK DESIGN
Master of Science Thesis in Industrial Design Engineering

A Lot with a Little
Evaluating Usability Methods for Developing a New Approach in Small Test Groups

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Luleå 5th of January, 2026
Isak Andersson
Viktor Sundström

Abstract

Since the late 1990s, there has been widespread discussion about how to test users and how many are needed for such a test. Nowadays, the standard is usually considered to be at least five users, a position based on multiple assumptions and misunderstandings. In this thesis, we explore this standard, along with the history of usability methods and a multitude of methods for evaluating everything from the user to the system to the method itself. After this, we define the user's experience and knowledge as the most important part of a user test, rather than the number of users. Given that the user is experienced enough, the number should not matter. The results are two new methods that can be used together with highly experienced users:
Specific Guidelines: A step-by-step method to create guidelines that define a project.
User-Centered Heuristics: where the user performs a heuristic walkthrough of a design, using their knowledge and experience to evaluate it.

Keywords: *Usability Methods, Heuristics, Method Creation, Method Evaluation, Design Theory.*

Sammanfattning

Sedan slutet av 1990-talet har det förekommit en omfattande diskussion om hur man testar användare och hur många som behövs för ett sådant test. Numera anses standarden vara minst fem användare, en ståndpunkt som baseras på flera antaganden och missförstånd. I denna avhandling undersöker vi denna standard, tillsammans med historien bakom användbarhetsmetoder och en mängd metoder för att utvärdera allt från användaren till systemet och själva metoden. Därefter definierar vi användarens erfarenhet och kunskap som den viktigaste delen av ett användartest, snarare än antalet användare. Om användaren är tillräckligt erfaren bör antalet inte spela någon roll. Resultatet är två nya metoder som kan användas tillsammans med mycket erfarna användare:

Specifika riktlinjer: En steg-för-steg-metod för att skapa riktlinjer som definierar ett projekt.

Användarcentrerad heuristik: där användaren utför en heuristisk genomgång av en design och använder sin kunskap och erfarenhet för att utvärdera den.

Nyckelord: *Användbarhetsmetoder, heuristik, metodskapande, metodutvärdering, designteori.*

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1 Introduction

The long-accepted standard within usability testing, spearheaded by the Norman Nielsen Group (Nielsen, 2000), is that for a method to be considered reliable, it needs a test group of at least five users. For some, fewer than five users are the norm, rather than the exception. We will begin with how we came about the problem with fewer than five users and what it evolved into.

1.1 BACKGROUND

Usability testing or Usability evaluation is a practice that has increased within the tech industry because its use is directly linked to users' satisfaction when new products are released onto the market. The goal of Usability evaluation is for the products to be easy to use, efficient, and effective in their application. A product achieves these metrics by eliminating usability problems, ranging from cosmetic to functional (Hwang & Salvendy, 2010).

Hwang and Salvendy (2010) examined which methods are commonly used in usability evaluation and noted that the most contested issue regarding the cost is sample size. The number of users or evaluators needed to find a given number of problems depends on how many problems each of them can find. Companies are using the sample size recommendation to argue for the most cost-effective way to increase usability metrics, and as a result, the importance of these numbers has increased, along with their use.

We, as engineering students, found ourselves relating to this way of thinking when we struggled to recruit participants for a usability test. Instead of focusing on cost efficiency, we worried about gathering enough user test data. We were told that the minimum number of users required for a usability test is 5, which made us wonder. Why is 5 enough? This number was first mentioned by Nielsen (2000), who claims that one could find ~84% of problems with 5 users. Lindgaard and Chattratichart (2007) state that this was adopted by many organizations as the optimum sample size for a usability test to yield a high return on investment, which explains why we had heard of this during our own studies. They continue by explaining that later studies found that reality was not as Nielsen had claimed it to be, and that his and others' research is based on multiple assumptions.

So why do companies and researchers care so much about these magic numbers? Well, Alroobaea and Mayhew (2014) describe that as the sample size increases, the reliability and the number of problems found also increase. At the same time, the amount of money spent proportionally increases. For this reason, companies often implement a low-budget iterative testing strategy where the importance of quantitative data is unnecessary. In other words, one shouldn't ask how many users are needed, but rather how many users can be afforded. And how many users do we have time for? And from that, what kind of user do we want?

1.2 OBJECTIVE AND AIMS

Based on the background, we sought out to explore usability methodology in order to create a method that situates itself as a low-budget alternative to already existing usability evaluation methods. To see the market's desirability for this type of research, we contacted and interviewed Saab because of their rise in the usability research department. From this, we constructed our main objective:

To evaluate existing usability methodologies and to develop a method that helps designers and researchers gather information with a low sample size.

Our aims are:

- To help designers and researchers perform usability evaluations when there are a few users available for testing.
- For one of our methods to widen the method landscape, for when organizations and companies are worried about not having enough resources to evaluate properly.

Our research questions are:

1. How can usability researchers interpret and prioritize usability problems while working with fewer than five participants?
2. What aspects of a digital interface can be evaluated without users?
3. How do other research fields mitigate the limitations of studies with fewer than five participants, and how could these approaches be applied to usability evaluation?
4. How can one avoid bias affecting the results when working with a small sample size?

1.3 STAKEHOLDERS

UI/UX Designers: Designers are the most affected, as the new methods can help them find answers to questions that were unavailable before. New methods and areas to further explore the field of usability, through a pair of new perspectives that can culminate in a better experience for their users.

Users: A group of users who have gone through some type of training or education and therefore possess specialized knowledge that can be used to evaluate products made for that knowledge. Our solutions are focused on this type of user, and bringing them these methods will affect how they will interact with their system of expertise when using our methods.

Product Owners: Responsible for overseeing a product, they often receive reports from the usability studies and make decisions based on them. Finding a solution that helps communication and representation of that data can help lower the risk for product owners. Our methods are in their interest because of the data they generate, and the analysis report provided by the researchers or designers.

Usability Researchers: Our findings are additional insights into what further research can accomplish in this field and what can and should be explored.

1.4 PROJECT SCOPE

As there is limited time to conduct validation tests, the results are what can be gained during the set time limit and are not to be considered the final iteration of our solution unless stated so. When using external feedback during tests, we are to use available participants to be able to conduct tests within the set timeframe. The project is not to be self-funded. While our goal is closely related to the budget, we will not have time to analyze the budgetary costs of performing our method and will have to rely on assumptions made based on other methods and interviews.

2 Context

Most people reading this document have an idea of what a usability issue is, but as explained by Tullis and Albert (2008):

"There´s no simple definition [for usability issues]…"

In this chapter, we look at the landscape of usability methods and their history. Additionally, we present our critique of Nielsen's Usability Heuristics.

2.1 USABILITY EVALUATION – A BRIEF HISTORY

Usability as part of human-computer interaction is a fairly new field. The evolution of the dialogue between humans and computers can be divided into three periods. 1950-1963 marked its beginning with the introduction of computers, with a period where usability or accessibility was rarely considered because users were the same as those who built the system, the programmers.

The period between 1963 and 1985 saw no change in the relationship between users and programmers. Instead, it saw an evolution of systems and models of interaction as companies wanted to increase efficiency, as the industry was leaving the Command Line Interface and instead adopted the Graphical User Interface. With the introduction of the personal computer came an increased interest to see how users thought about the new interfaces, which marked a shift from the primary testing method introduced in early 1980s: laboratory user testing (Federici & Borsci, 2010), and towards a more accessible way to test user understanding with the origins of usability evaluation methods like think-aloud, introduced by Lewis (1982), and the System Usability Scale (SUS) by Brooke (1986).

After 1985, usability and accessibility became the centre of interaction design, focusing more on usability issues with the rise of the personal computer and the internet. This came with the consequence of separating the user and the programmer, beginning the first need for proper user testing (Federici & Borsci, 2010). By 1990, developers explored other evaluation methods to reduce the cost of testing compared to traditional methods. Up until this point, testing was performed late into the process, so there was also a need to find methods to be used earlier with prototypes (Federici & Borsci, 2010). This period introduced many expert-based methods that utilize guidelines and avoid the user, which are still popular to the present day[1], with two examples of methods introduced during this time being heuristic evaluation by Nielsen and Molich (1990) and Cognitive Walkthrough by Lewis, Polson, Wharton, and Rieman (1990).

After the 2000s, technological innovation skyrocketed, and methods for usability evaluation are more important than ever. There are multiple methods used daily, with the some of the most popular ones summarized by Rohrer (2022):

- Usability testing
- Field studies
- Contextual inquiry
- Concept testing
- Diary studies
- Customer feedback

[1] This thesis was written in 2025.

- Participatory design
- Focus groups
- Interviews
- Eyetracking
- Usability benchmarking
- Remote moderated testing
- Unmoderated testing
- Desirability studies
- Card sorting
- Tree testing
- Analytics
- Clickstream analytics
- A/B testing
- Surveys

2.2 A CRITIQUE OF NIELSEN'S USABILITY HEURISTICS

Nielsen's Usability Heuristics (1994) are considered a standardized method in and of themselves to the point where people associate Heuristic Evaluation with Nielsen's guidelines. This has put the method under much scrutiny, which has only strengthened its status. In this chapter, we discuss the problems that come from the standardization of Nielsen's Usability Heuristics. To begin, a major limitation of Nielsen's Heuristics is that it neglects the users. According to Nielsen and Molich (1990):

"[Heuristics is] looking at the interface and passing judgment according to one's own opinion."

One could use the ten Heuristic Guidelines to create a specific user test, but this would constitute a different method than Heuristic Evaluation. To evaluate the applicability of Heuristics Evaluation, one must consider what is lost when users do not partake in the evaluation process.

Conceptual models describe how an object is expected to function, which is conveyed through its usage and shape. Norman (2002) divides this aspect into two parts: The designer model, which is how the designer expects the user to handle the product, and the User's model, which is the mental model the user creates through interaction with the design. Both of these are then compared to the System Image, which is the actual product itself. As an interesting note, this carries similarities to Peirce's Theory of Signs (Peirce, 1932), where Pierce argues that any object carrying information is made up of 3 parts:

- **Representament**: What the sign is meant to represent.
- **Object**: The actual object itself.
- **Interpretant**: How the sign is interpreted.

In this case, the representament could be interpreted as the designer model, which corresponds to the designer model that the system (or a component of it) is intended to represent. The object would then be the physical design, with the interprant being the User's model. One could even compare this to the Criterion concept mentioned by Hartson, Andre, and Williges (2001), where it is the company's goals of the product versus the designer's goals to complete the product, called Ultimate- respective Actual- Criterion. These two overlap at one point called the Criterion Relevance, which is what each party actually wants to accomplish, in this case being the final product. The point here is that there are multiple theories about how we perceive the same object or concept differently.

The designers expect the user's model to be equal to the designer's model, which is

not necessarily true. In fact, it may be the complete opposite, which is why Norman highlights the importance of how the design is presented.
Adding to this, an effective way to see how the user's model aligns with the designer's is through user testing, where the user's model is examined and evaluated by the designers. This allows the designers to then model the system image to better fit with the user's expectations. This follows the findings of Hertzum (2006), where several usability evaluators stated that they needed to see users interacting with the system to assess the severity of usability issues.

In the original document by Nielsen and Molich (1990), it is noted that the Heuristic Guidelines are general, stated as:

"These nine[2] principles correspond more or less to principles which are generally recognized in the user interface community, and most people might think that they were "obvious" if it was not because the results in the following sections of this paper show that they are difficult to apply in practice."

The important part here lies in the usage of "generally" and "obvious". The implication is that all these guidelines are understood within the interface community, but are never stated obviously. This explains why such guidelines are created: to make these otherwise implicit assumptions explicit to the wider community. But as is the problem with any guideline intended to apply to a larger group, certain aspects of it will become harder to apply to certain designs. This could be attributed to difficulty in understanding the guidelines, something that is stated in the original text by Nielsen and Molich (1990). Alternatively, because the guidelines are general, the usefulness of individual heuristics may vary between applications.

With all of this in mind, we proceeded with an interview with UI/UX designers, where we asked them about their usage of these guidelines. In response, they explained that out of the ten guidelines, there were two they found hard to apply to their own project. This would imply that this theory holds up to a certain degree.

[2] The 10th guideline was added later by Nielsen in 1994.

3 Theoretical framework

This chapter presents Industrial Design Engineering and how this field relates to this thesis project. We will also present the theory that was used to structure our user tests.

3.1 INDUSTRIAL DESIGN ENGINEERING (IDE)

Industrial Design Engineering (IDE) is grounded in design thinking, as laid out by Brown (2009). Design thinking aims to join overlapping spaces of innovation to solve a problem by accepting the project's constraints. To aid in innovation, IDE practices exploration in support of fundamental human needs using technology through consideration for a project's feasibility, viability, and desirability. This process, seen by others as messy, unorganized, and extending the product development timeline, is an enthusiastic exploration that, when done by an experienced design thinker, allows the design to be adjusted through agile practices, avoiding predictable and unproductive patterns (Brown, 2009).

This discussion continues by examining the designer's agility and its connection to decision-making ability. For example, if we were to imagine a designer in a vacuum, we might wonder how the designer could evaluate and decide on a design independently. We could say that what the designer ultimately decided on was based on a feeling; however, that feeling has a name: intuition. Since designers are human, it is reasonable for them to make human-like assumptions about how their design will be used (Milton & Rodgers, 2023). Designers also have the benefit of gaining a deeper understanding of a person's thought process. Therefore, it is not unreasonable for the designer to make certain assumptions about a design. In IDE, designers use intuition as a method when they have processed the data from users, theory, and observations. The understanding of these perspectives is very important to verbalise when presenting why a decision was made; additionally, it can be referred to as a method of design, as per Milton and Rodger's (2023) description.

Design methods have been used since the beginning of design. However, they were often undocumented and applied internally, similar to intuition. Design thinking, as discussed by Ford and Yoho (2024), supports the idea that design uses methods, stating that design thinking is a comprehensive research method presented as a five-step problem-solving process. Using methods is essential for distancing the designer from the work, as reliance on personal assumptions and values risks shifting focus away from users. Methods are tools to understand different perspectives, from the well-known method of brainstorming (Wikberg-Nilsson, Ericson, & Törlind, 2017) to those used in a specific field of design, such as Heuristic Evaluation (Osvalder, Rose, & Karlsson, 2019).

Wikberg-Nilsson, Ericson, and Törlind (2017) argue that being process-oriented is one of the important attributes of a designer, focusing on alternative solutions and examining design problems from a holistic perspective. In IDE, it is important to have an iterative approach and a focus on the process to be able to confidently contribute to a satisfying user experience.

3.2 HOW MANY USERS ARE ENOUGH?

There are numerous magic numbers in usability, referring to the minimum number of participants required for testing. The problem with these comes from the fact that they vary widely, ranging from 5 in usability studies (Nielsen, 2000) to 20 in usability metrics (Nielsen, 2012), with variations such as 10±2 for usability studies (Hwang & Salvendy, 2010). Tullis and Albert (2008) discuss this further and compiled several cases where the magic number was 5 and cases where the magic number was anything but 5. This raises the question of where these magic numbers come from. In general, the magic number of users can be calculated using methods described by the authors mentioned above using a simple binomial distribution, as described by Nielsen and Landauer (1993):

$$1 - (1 - p)^n$$

Where p is the generalized percentage of problems found by a single user and n is the number of users evaluated. In the case of Nielsen's minimum amount of 5, he used a p-value of 0.31, which gives us Figure 1:

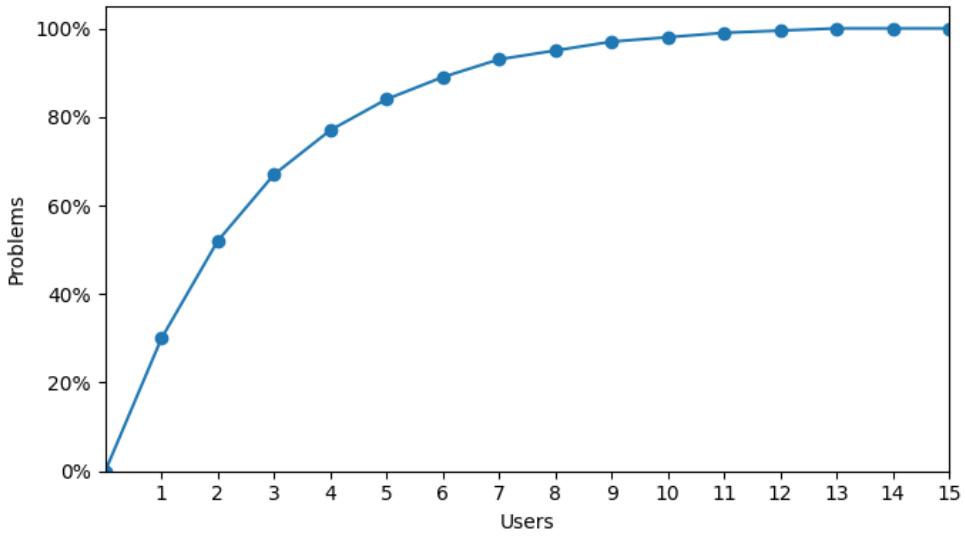


Figure 1: Number of problems found by the number of users. Authors' own illustration.

Note that the graph itself shows that reaching approximately 100% problems found requires 15 users. Nielsen himself notes this and states that while testing with 15 users is recommended, it's better to split it up into three smaller tests. This is where the 5-user minimum originates, as Nielsen's example was interpreted as a recommendation rather than an illustration. Furthermore, Schmettow (2012) argues *against* the usage of minimum numbers, as there are too many variables of user testing that are ignored when using the equation above, to the point where it loses functionality. This exemplifies the problem of relying on fixed magic numbers without accounting for contextual variables.

The function of these magic numbers can be examined by considering the effects of using too few or too many participants in a user test. Nielsen (2000) argues that, outside of wasting resources, too many users bring in very little new information, as the importance of a minimum lies in how resources can be allocated optimally to gain

information about the system. He then points out in the same article that more users are required when there are several distinct groups of users. Nielsen also states that three users are enough to get a good idea of the diversity in user behaviour. This follows our interviews, where experts[3] stated that three is enough in certain contexts. Hwang and Salvendy (2010) argue that this is incorrect, with five users only being able to find simple problems at most, and 8 to 12 being able to find severe problems. This point is interesting, as it argues more for the severity of the problems themselves rather than the amount, an important point when it comes to user testing.

As another argument, Molich et al. (2004) tested consistency among different testing groups, consisting of various amounts of experience among the users, and stated that;

"It appears that five test participants are enough to find worthy problems, but that one test is, by itself, unlikely to find anywhere near 85% of the problems regardless of the number of test participants. It may very well be true, however, that for a given test team and a given set of tasks, testing more than five users will not reveal a significant number of new problems."

In other words, they argue that the problem lies not in the number of users but rather the number of tests done on a system.

Kirakowski and Corbett (1990) continue talking about small sample groups, this time from a statistical point of view. Here, the problem lies in situations with far too few users, or rather, small population groups, which make it difficult to create assumptions about the underlying population parameters. This, in turn, may cause a situation where statisticians must rely on chance, which is generally considered inadequate in a scientific context. Because of this, non-parametric analysis is used instead in these situations, where differences between groups can be compared without any assumptions about the statistical distribution of the data.

If we want an even more extreme case, there has been research with a sample size of one within the realm of psychology. Dukes (1965) studied a multitude of single-subject studies and noted that the subjects of these studies can be split into two types: the first type being case studies, where it is typical to do an in-depth analysis of an individual, and where more knowledge about the subject is required. The second type of report is done on the basis that the individual being observed is so general that they can apply to any user within that context. Generally, within a design context, it is difficult to determine whether a case is sufficiently unique or sufficiently general to justify a single-user test. One such example is a product designed for a specific individual, such as a prosthesis. Even in this scenario, it highlights the constraints involved in defining an absolute minimum number of users.

[3] See Chapter 4.2.1

3.3 IMPORTANCE

Usability tests generate and amass insights in the form of problems and concerns, for example, as Dumas and Redish (1999) state, the designer will need to find a way to organize them, and that importance is one of the ways to organize usability findings. Dumas and Redish divide importance into two dimensions:

- Scope: How widespread is the problem?
- Severity: How critical is the problem?

In this chapter, we present the findings on these two dimensions and how it applies to user testing.

3.3.1 Scope

Dumas and Redish (1999) mention global and local problems and emphasize the scope rule: global problems are more important than local problems. A local problem is often contained on one screen, dialog box, window, or manual page, while global problems have a scope larger than a single component. If it is found during a user test that a dialogue box is confusing, it is a local problem. If it is later revealed that this is due to a design flaw that exists in every dialogue box, then it is a global problem. They continue by mentioning that those who are new to human-computer interaction, and many designers, tend to misinterpret symptoms of global problems as local issues. When trying to address the local problems, Dumas and Redish (1999) state that seeing the connections between them is difficult and something that comes from experience or training. Similarly, if a problem is treated as a local one while it is global, it may not fix the problem itself, leading to wasted time, energy, and resources in many cases.

3.3.2 Severity

Knowing how widespread a problem is will still not explain which issue should be fixed first, which is where severity comes in. Sauro (2014) mentions that there is no correlation between the number of times a problem might affect a user and its severity. Hertzum (2006) discusses this further, mentioning that severity can be defined through usability experts evaluating the system. Tullis and Albert (2008) describe systems that base severity on user experience, usually by rating the problem with a scale from low (frustration or annoyance) to high (task failure). They continue by mentioning that critics of basing severity on experience believe that severity should be fully left up to the user, and mention that there is little agreement between evaluators of what constitutes a serious problem.

3.3.3 What does this mean for User Testing?

The implication of this is that it may be easy to identify problems in a system, but the difficulty lies in verifying their severity and, therefore, determining how project resources should be allocated. This is where user testing becomes important, as involving a sufficient number of users provides a clearer indication of a system's problems and their severity, as shown in Figure 2. A major problem with this approach is that these magic numbers vary widely between applications, meaning that even then, one can't guarantee that enough data is gathered without a deep analysis.

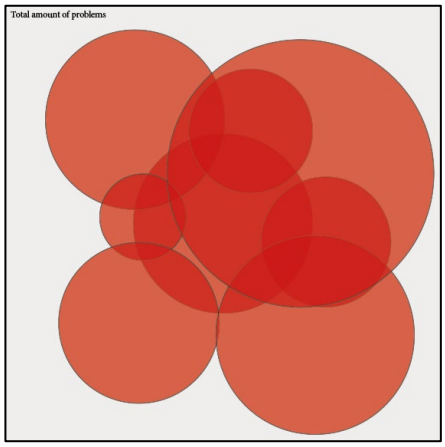


Figure 2: A lot of user test participants give a good idea of the number of problems and their severity. The red represents users, while the square represents the total number of problems.

If it is not possible to recruit enough users, the issue shifts away from identifying the number of problems in a system, illustrated by the area covered in Figure 3 (left), and instead toward verifying their severity, represented by the percentage of overlap in Figure 3 (right). This makes it difficult to guide resource allocation or, in the worst cases, to determine which solution should be used to address the problem.

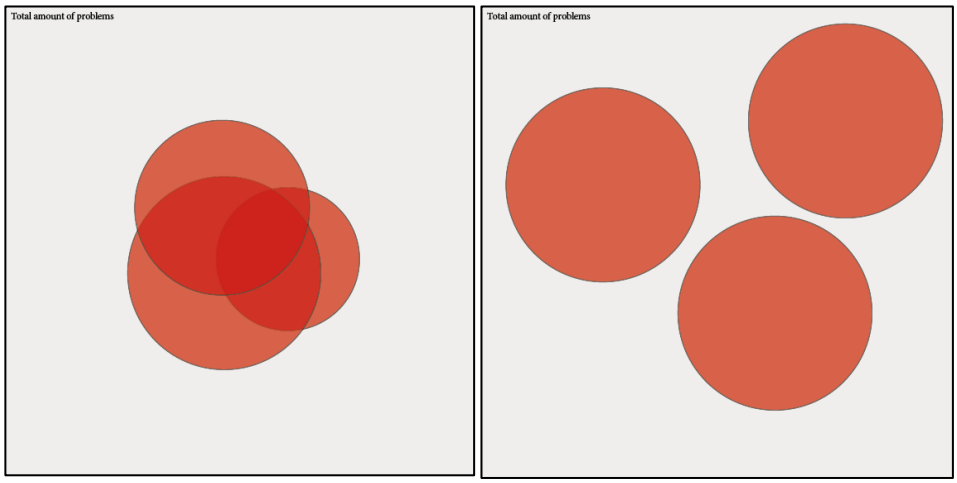


Figure 3: If you have too few users, it will be hard to tell the severity or scope of the problem itself.

To summarize, when talking about the importance of problems found in a system, it is less about identifying the number of problems and more about ensuring that the problems discovered are significant and/or widespread enough to justify allocating resources to fix them.

3.4 METHODS FOR EVALUATING

As this project explores how to evaluate usability testing conducted with a few users, it is important to understand how the current methods are used. While most of these methods were not used in this project, it is vital to understand how one can evaluate user behaviour, design solutions, or evaluation methods.

3.4.1 Methods for Evaluating a System Using Participants

User testing involves observing participants and measuring their performance by different measures, such as on-task completion, ease of use, or their perceived experience of the system. Formal user testing is typically conducted in a laboratory to ensure a controlled environment. Informal user testing is often conducted in the system's implemented environment, where it is normally used. The way a user test could be performed changes depending on what type of information you care about, which is based on what is most appropriate for the situation (Kock, Bijon, & Pretorius, 2009).

Kock, Bijon, and Pretorius (2009) discuss eye-tracking, focusing on how information is tested, determining effectiveness, efficiency, and user satisfaction when compared to Heuristic Evaluation in terms of problem detection. They concluded that user testing is effective at finding usability problems that are knowledge-based, and is not the most effective at finding information and usability problems that are skill-based.

Think-aloud (Lewis, 1982) is a popular method where a user explains what they are thinking while performing tasks. This provides direct insight into how the user not only acts but also feels during the tasks. It also reveals insight into how the user views the design itself, as illustrated by an example from Lewis and Riemann (1993):

> *"The word 'parameter' was used [a lot] to refer to a quantity that could take on various user-assigned values- Test users read this word as 'perimeter', a telling sign that the designers had stepped outside the vocabulary that was meaningful to their user."*

It is not uncommon to record think-alouds for later review, as it can be easy to miss subtle visual cues in the moment, such as the example described above. The main strength of think-aloud lies in its ease of implementation and effectiveness in gathering significant amounts of qualitative usability data. The danger comes from taking users' wishes too literally, as Nielsen (2012) notes from the beginning of the document:

> "*Listening to what people say is misleading; you have to watch what they actually do."*

Another popular method is the System Usability Scale (SUS) (Brooke, 1986), which is a paper with a few questions that a user gets to fill in after a system evaluation. This paper has been tested and evaluated many times and has proven effective for measuring overall user satisfaction, providing information about user opinion on the system. Like think-aloud, it is easy to implement into any user test. The main problem is that SUS does not provide detailed information about specific usability problems or what to fix, as it only gives a value of the user's opinion of the general system (Klug, 2017).

Interviews as a method come in three common variations: unstructured, semi-

structured, and structured; however, we will not focus on the Unstructured interview as it is less common in usability research. The other types of interviews share predefined questions, with the difference between semi- and structured being the open-endedness. A structured interview is easier for inexperienced interviewers to do and can be conducted face-to-face, over the phone, or during a video call, and it produces more consistent responses compared to a semi-structured one. In comparison, the semi-structured interview must be performed by someone who has knowledge or experience in questionnaire design for it to be effective. The strength of it is that it can address complex issues that a structured one may miss, and it can be outsourced to a consulting company because of the limited base set of questions. While it requires less preparation than a fully structured interview, there are risks due to differences between interviewers and a need for consistency between them if multiple interviewers are to perform that interview. If not, researchers risk collecting data that is very hard to analyze properly (Wilson, 2014).

3.4.2 Methods for Evaluating a System without Participants

Before performing a user test, a designer might perform a Heuristic Evaluation, as described by Nielsen (1994). At its most basic level, it involves examining a design and forming judgments about its usability. More commonly, the system is compared to a set of guidelines, most notably Nielsen's Usability Heuristics (1994), which list standardized usability features. This method is used to identify common usability mistakes in a system. While useful, it requires a designer trained in performing this method, as without sufficient knowledge of design or the type of system being evaluated, the evaluator's opinions might not be applicable[4]. This is partly why guidelines are used, but they must be well understood before they can be properly applied to a design. These guidelines include the following principles:

1. Visibility of System Status
2. Match Between the System and the Real World
3. User Control and Freedom
4. Consistency and Standards
5. Error Prevention
6. Recognition Rather than Recall
7. Flexibility and Efficiency of Use
8. Aesthetic and Minimalist Design
9. Help Users Recognize, Diagnose, and Recover from Errors
10. Help and Documentation

Another set of guidelines is Metaphors for Human Thinking. Hornbæck and Frøkjær (2002) compare it to Nielsen's Usability Heuristics (1994), but rather than standardized usability features, a system is compared to metaphors for human thinking, which explains its name. It presents a similar limitation where the designer who wants to use this method must be trained in the method, or they might misunderstand the different metaphors. The metaphors are as follows:

1. Every person is like a landscape eroded by water.
2. Human thinking is like a stream of water
3. The mental activity is like a jumping octopus in a pile of rags
4. A person's utterances relate to the person's insights as the splasher over the waves to the rolling sea.
5. Human knowing is like a site of buildings in an incomplete state of

[4] Together with what is mentioned in Chapter 2.2.

construction, developed through maintenance and rebuilding.

There are also Cognitive Walkthrough (CW) and Predictive Human Error Analysis (PHEA), as described by Osvalder, Rose, and Karlsson (2019), in which designers conduct a structured, systematic analysis of an interface by putting themselves in the user's position. Like Heuristic Evaluation, they use questions to help the designer adopt the user's perspective. While Cognitive Walkthrough was developed for analysing single technical products with a screen-based interface, it can be used for products without a graphical interface as well as for evaluating work tasks in control room environments (Osvalder, Rose, & Karlsson, 2019). One or more evaluators with knowledge of cognitive ergonomics, interface design, and risk analysis can use these methods, with some domain knowledge linked to the use case of the interface. Real users can also be included in the evaluator group, but need to have knowledge of how the product and its interface function, as well as the circumstances under which it will be used. Throughout the process, Osvalder, Rose, and Karlsson (2019) describe that the evaluators play the role of the user and actively try to find problems with the interaction and issues that may occur. They also divide the analytical process of CW and PHEA into four main steps, where each step can be performed separately:

- Definition of the evaluation – Deciding the framework
- Description of the human-machine system
- Analysing
    - Whether the user will act correctly– CW
    - What errors the user can make – PHEA
- Summarize and follow up on results

As another method to view the system from the user's perspective, Lewis and Rieman (1993) also describe Action Analysis, in which the steps taken by a user are defined. These steps are then analyzed to estimate the time required for a user to complete a task. To better estimate the time it takes to perform each step, they created a table[5] based on averages from hundreds of user tests, although researchers could do this themselves, provided that data from previous user tests have been retained. There is an alternative version of this method in which, instead of evaluating individual steps and their duration, one examines the overall task structure, focusing on the actions required to complete the task and whether the task can be completed efficiently in this manner.

Hornbæk (2010) has also discussed some of the widespread problems when it comes to evaluation in general, with one being that the evaluation of problem value is often omitted in most texts discussing methods. Value in this context refers to how important that problem is in the context of the system itself. Another mistake mentioned is that the method itself is treated as definitive, which is to say that problems are often attributed solely to the method. This assumption ignores the role of the user, as it is fully plausible that the user found a problem through experience with similar systems or through accidental user behavior.

3.4.3 Methods for Evaluating Users

User tests are the most common method of evaluating a system with users, but how they are conducted can vary significantly between designs. Many factors may affect users during testing. One such factor is bias, which influences many evaluations.

[5] Due to its size, it will not be presented in this document.

There is, of course, a movement to analyze and categorize these biases in different contexts. Arnott (2006) compiled a list of cognitive biases that have proven themselves to be persistent in laboratory studies[6].

To address these biases, he refers to Keren (1990), who defined a framework to debias, inspired by medical studies. First, one must identify the existence of a bias, then one considers how to reduce or eliminate the bias, and finally monitors the effectiveness of the intervention. Fischhoff (1982) also mentioned a method to overcome some cognitive biases. The method primarily involves informing users about these biases and later training users to avoid them. In short, there are no direct methods to prevent bias, but rather methods that aim to mitigate bias through increased awareness.

Following this, Lash et al. (2014) discuss when it is fitting to analyze biases in each context. They discuss that bias analysis is not needed if the discussion is kept on the methods and data, but once it is connected to specific results, bias may influence the assumptions underlying the results. Due to this, the data used for decision-making must be analyzed for potential bias. It is mentioned that a simple bias analysis might suffice in this case, but some level of analysis remains necessary. Snyder (2006) argues that bias cannot be completely removed but should be acknowledged to alleviate its effects. Tullis and Albert (2008) distill the different sources of bias into six categories, following the argument of Snyder:

- Participants, the users you are testing
- Tasks, how you've formulated the tasks
- Methods, the methods you use to evaluate
- Artifact, the nature of the prototype or product you are testing
- Environment, the physical area where you are testing
- Moderators, the ones who are observing the test

In addition to bias, mental workload is an important factor, especially when it comes to interface design. One of the more well-known methods is NASA-TLX. Osvalder, Rose, and Karlsson (2019) describe it as a subjective method in which six dimensions are evaluated on a scale from 1 to 100, where 1 is no mental load and 100 is an extremely high mental workload. The six dimensions are as follows:

- Mental Demand
- Physical Demand
- Temporal Demand
- Performance
- Effort
- Frustration

Another method mentioned by Osvalder, Rose, and Karlsson (2019) is the Subjective Workload Assessment Technique (SWAT), which uses three dimensions scored from 1-3, with 1 being low mental workload and 3 being maximal. The three dimensions are as follows: Psychological Stress Load, Mental Effort Load, and Time Load. Zahra (2024) compares these two against each other and evaluates that, while NASA-TLX is widely accepted and works well, SWAT is easier to use while also being less detailed in its results.

[6] This list will not be published in this document due to its size. If you are interested, we urge you to read the original document.

3.4.4 Methods for Evaluating Methods

As this project focuses on evaluating methods, it is important to understand how different methods themselves can be evaluated.

Hartson, Andre, and Williges (2001) experimented with a method of evaluation, focusing on its ability to identify problems and how these problems should be understood within the method. They mention three core characteristics for evaluating a method:

- Thoroughness, referring to the ability to find problems.
- Validity, referring to the relevance of identified problems.
- Reliability, referring to the consistency of results across users.

Siau and Rossi (2011) instead proposed a set of categories for classifying methods. One important aspect of their work is what they call "*Theoretical and Conceptual Investigation*", where a method is evaluated using another model, such as a statistical model. Another category they identify is "*Empirical Evaluation*", where empirical data are collected and measured. such as a field experiment, in which the method is applied directly to participants in practice.

3.5 USERS' EXPERIENCE

Because this thesis is based on the argument that experience is more relevant than the number of users, it is important to understand how user experience affects user testing.

3.5.1 Trained vs Untrained

As one would expect, there is a difference between evaluating trained users and untrained users, giving different types of information. Kjeldskov, Skov, and Stage (2010) studied some of the differences and concluded that untrained users tend to perceive issues as more severe. They also mention a notable finding about the number of usability issues identified, which is that an untrained user might find more issues than a trained user. The main point being trained users identify more critical issues, while untrained users identify a broader range of general issues. Our interviews also indicated that trained users allow a lower minimum number of users for the number of users needed for a test, with the implication being that more specialized users are more effective test participants.

To examine this further, we can return to the binomial function from Nielsen and Landauer's (1993) study[7], which represents the number of problems found based on the number of users. The number 0.31 is the percentage of issues found by a single user, which is an average they created from other user tests. Now, this doesn't mean that the value is incorrect, but rather that it works on the assumption of the users themselves. We don't know the experience of the users behind those values[8], and we can assume it is likely a mix of trained and untrained users that gave us that number. This raises the question of how the value would change when using only experienced users. Based on our interviews, we can infer that the percentage of problems found would be larger. As noted by Kjeldskov, Skov, and Stage (2010), experienced users would not identify more problems, but those problems would be more *severe*. This makes it harder, as it implies that we cannot use the binomial function to argue for how many experienced users are needed to find issues. We would also need a function

[7] Shown and further discussed in Chapter 3.2.

[8] With an important note being that there is a high variance between those different tests; sd = 0.12

to quantify the number of critical issues identified to properly evaluate that situation. This aligns with Schmettow's study (2012), where either more variables need to be thought of when using this binomial function, or another method is needed entirely.

3.5.2 Co-Design

The belief that everyone is creative has historically not been widely accepted; however, Sanders and Stapper (2008) state that design has changed from its traditional form. They continue by stating that the design process is shifting toward greater emphasis on understanding the future user, with one way to practice it being co-creation. Described as the act of more than one designer working together in the design process, the co-design aspect of co-creation treats everyone as a potential design partner. In co-design, compared to traditional user-centered design, the role of the user is the main shift. Involving the user earlier in the process makes their role more significant as experts in their individual experiences and knowledge. The designer is still important and has a slightly different role, more of a facilitator, responsible for creating design tools necessary for the users to contribute effectively (Sanders & Stapper, 2008).

Zamenopoulos and Alexiou (2018) look at it from another perspective, describing Co-Design as allowing anyone to partake in the creative process, no matter their prior training. Zamenopoulos and Alexiou (2018) provide an example of a health organization gathering its mentally unstable patients to find a new solution to improve the services provided to them, or a small community that collaborates with academics and the third sector government to find a way to help socially isolated elders. Both of these carry the focus of a shared goal, but uncertainty arises regarding whether the shared goal is clearly defined. Together with this, there is an implication of the importance of the user's knowledge about the subject; a mental health patient has knowledge and understanding about how they are being treated (outside of extreme cases), and members of a small community might have personal experience about socially isolated elders within the community.

Taken together, these perspectives suggest the user being an expert in a very specific subject, that instead of a designer with experience in design, is a designer with experience in the subject. While Zamenopoulos and Alexiou (2018) state that co-design can be performed with users without any prior training, some level of training could form part of the user's experiential contribution.

3.6 STATISTICAL EVALUATION THROUGH USER TESTS

Kirakowski and Corbett (1990) state that one should define a clear objective before conducting tests, as this mission defines which methods should be used for evaluation. In addition to this, variables can be defined to further constrain the choice of evaluation methods to evaluate.

Variables can be defined as qualitative or quantitative, such as a severity scale. Tullis and Albert (2008) discuss converting qualitative data into quantitative measures for statistical evaluation and comparison across datasets. When discussing converting between qualitative and quantitative data, they mention the think-aloud method, as well as recorded sessions where evaluators did not alter their tone or facial expression, along with possible explanations for why such changes occurred.

The reason these are recorded lies not only in the mood but also in the surrounding

context. A quantitative value could instead be derived from the number of mood changes from positive to negative states throughout the tasks. This results in two values: a qualitative value representing context and a quantitative value representing frequency. This raises the question of the purpose of measuring the number of mood changes, to which it can be argued that one would arguably not want too many of these types of mood changes to happen while using the system.

4 Method and Implementation

This chapter describes the structure of this project, together with the methods used and how they were used to develop and implement the solution.

4.1 PROJECT FRAMEWORK

The project followed the double diamond methodology described by Brown (2009), which divides a project into a problem space and a solution space, each consisting of a diverging phase and a converging phase. These phases were defined for this project as follows, seen in Figure 4:

- **Exploration**: Where we analysed the problem, explored similar solutions, and searched related theories to gain an understanding of the relevant problem space.
- **Ideation**: Where we used different creative methods to generate ideas for addressing the problem.
- **Implementation**: Where we defined a design concept and complemented it with theory, and structured evaluation tests to validate the concept.
- **Refinement**: Where we evaluate our test results and refine the concept and its presentation into its final form.

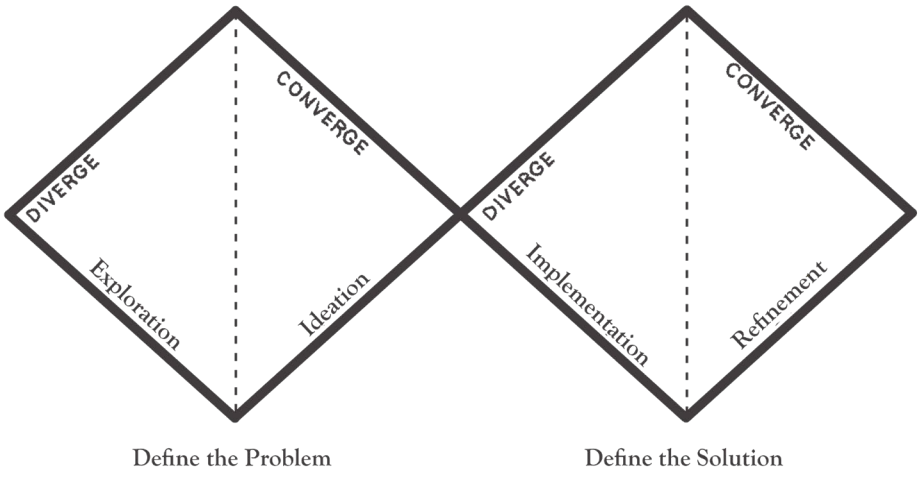


Figure 4: Double Diamond methodology outlining the project. Author's own illustration.

Each of these parts defines half of a diamond, with Exploration and Implementation being diverging phases intended to widen our knowledge. Ideation and Refinement are converging as these phases are to narrow the exploration to determine which direction to pursue.

4.2 EXPLORATION

For the Exploration phase, the main goal was to explore the context and central research question of the thesis. To understand the problem, we used expert interviews, a literature review, and meetings with stakeholders. Part of the literature review is shown in Figure 5.



Figure 5: Studying Literature. Autor's own Photo.

4.2.1 Interviews

To gain a better understanding of how the problem is viewed by professional designers, three subject-matter experts participated in a short interview where we questioned: "If you had to evaluate a design with a maximum of three subjects, how would you go forward?"

In addition, we interviewed three usability researchers with three usability researchers working in the industry, together with their UX director. Together with this, we performed several short interviews with different UI/UX designers in the industry throughout the project. Each participant requested anonymity and will be referenced generically as experts where appropriate.

During the semi-structured interviews with the usability researchers, we focused on how they perceive usability testing methodologies and how it changes as the number of available test participants decreases. We also asked questions about whether they had any example studies they could describe in detail. The full interview guide can be found in Appendix 1.

4.2.2 Literature Review

A literature study, as described by Snyder (2019), was conducted primarily through searches for articles in Google Scholar, Science Direct, ACM Digital Library, and

Scopus with these keywords: evaluation design, UX/UI design evaluation methods, evaluation methods, user testing, and UX evaluation. To reduce selection bias, we also explored articles from specific journals, including the International Journal of Human-Computer Interaction, the International Journal of Design, and similar journals. Articles focusing on user evaluation and method evaluation were selected. Articles were included only if they were written in either English- or Swedish-language publications.

4.3 IDEATION

During the Ideation phase, we apply our newfound knowledge to create concepts, as shown in Figure 6.

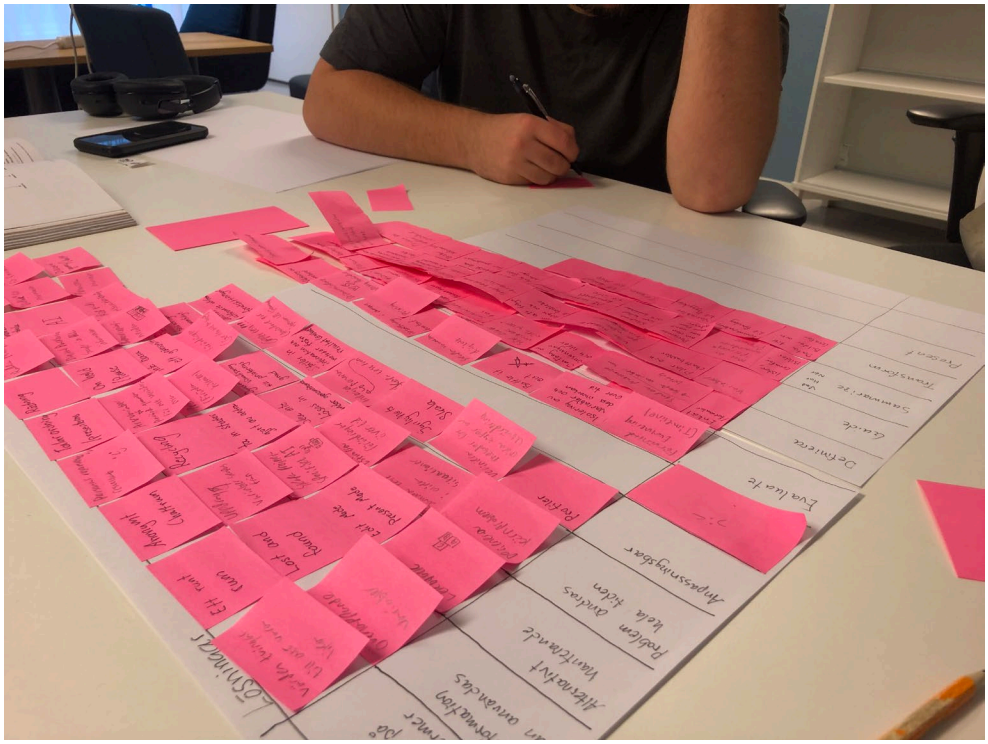


Figure 6: Creative Work. Author's own Photo.

4.3.1 Brainstorming

A brainstorming session was conducted to generate ideas. To better structure the session for this project, the Walt Disney method was used as described by Dilts (1995), which divides the ideation phase into three parts. With a focus on the question "What could help when there are few to no users available to perform user tests on for a project", we did the following:

- The Dreamer, which we used as a brainwriting (Wikberg-Nilsson, Ericson, & Törlind, 2017) session, where we wrote down as many potential solutions as possible.
- The Realist, where we attempted to apply the generated ideas to real-world contexts.
- The Critic, which we used to assess the viability of each idea in the context of our project, to determine whether the idea was suitable for further development.

For our project, we stopped after two iterations, as it became clear that we had focused on a few specific ideas created through this method.

4.3.2 Analogies

Subsequently, analogies were used, as described by Wikberg-Nilsson, Ericson, and Törlind (2017), to further expand idea generation. Here, several figures and concepts are used to create analogies related to our project. The analogies and their interpretations were as follows:

- Trial

An area where two different positions argue against each other. Ultimately, a decision must be made, independent of the feelings of each party.

- Pippi Longstocking

Independent and strong, she does not follow the standards of society. It could be an idea that is different from what is normally expected.

- MacGyver

A character from an 80s television show, where he became famous for escaping any situation by using anything available at the moment.

- Fox Pit

Fox pits are said to have two exits, in case it needs to escape. It could be an idea that allows multiple interpretations or is maybe less risky.

- Meet the Robinsons

A quirky family, full of colourful characters. It could be a solution that allows anyone to make it work for them, or maybe multiple different ways to use it.

- Iceberg

It is well known that the tip of the iceberg, the only part visible, only represents less than a fraction of the whole. It could be that only a small part of the solution can be viewed by the user, while the rest happens in the background.

- DNA Molecule

Something that contains a lot of information, connected within the molecule. More molecules can be connected to create strings. This can refer to the connections needed between different ideas.

- Frank Sinatra

A famous singer who sang "I did it my way." It could allow the user to do something at their own terms, regardless of constraints.

- Tom & Jerry

An old cartoon about a cat hunting a clever mouse. It could represent an eternal chase, a goal that can never be completed but could be strived for.

- The Addams Family

An old TV show about a family that stands out from normal society, generally by being the opposite of what is expected. This could refer to acting the complete opposite of what others expect you to.

- Katjakaj & Bentibent

An old Danish kids' show about two friends doing everyday things. As one of the aspects that was remembered the most among us was that they would watch a TV painted on the wall. It could represent maximizing outcomes with minimal resources.

- Schrödinger's Cat

A thought experiment that presents a cat that can be considered both dead and alive at the same time. It could represent something that is in two states at once, or the unknowability of certain actions or ideas.

These analogies were then discussed in relation to how they represented the project, both in terms of the goals that we should strive for and how people act towards our problem. This creates a list of objectives and ideas that apply.

4.3.3 Creative Matrix

The Creative Matrix was inspired by the Morphological Matrix method. As described by Wikberg-Nilsson, Ericson, and Törlind (2017), it is primarily used to identify partial solutions by defining problems within a product. Through a function analysis, a product's criteria can be found. The problem is that we are developing a system or method, instead of something physical, which requires an alternative approach to defining criteria. Our criteria were derived from the project objective and discussions with researchers. In the end, they became:

- We want to help **define** the problem, so we can create a method that solves it.
- We also need a **guide**, as the method either needs to be clear enough, or we need to find a method to help understand it.
- The method should **summarize**, as part of the mission is to show information clearly.
- Like the above, **transform** was defined because we also need to convert the information into a legible form for stakeholders outside the project.
- And finally, **present**, as we are interested in helping present this to someone outside of the designer group.

In addition, analogies from the previous phase were used to expand our thinking. For each criterion, we had a 5-minute brainstorming session, and after all the criteria had been addressed, we combined the criteria into concepts.

4.4 IMPLEMENTATION

During the implementation phase, we focused on verifying our theories and concepts with five think-aloud user tests. A part of one of these tests is shown in Figure 7.

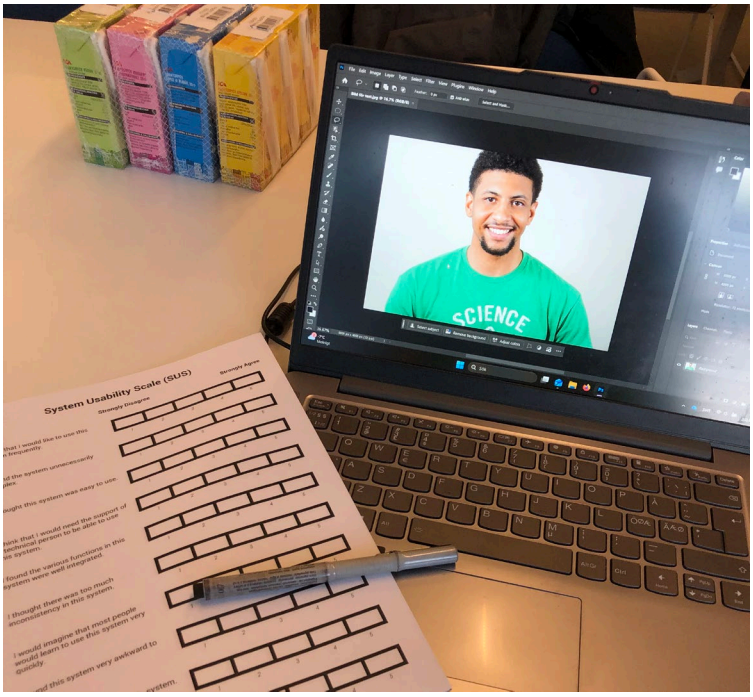


Figure 7: Testing the Concept. Author's own Photo.

To simulate historical data, the test was structured to contain a pre-test and a comparative test with two similar programs. The two tests are the same, outside of being performed on different programs. As most users available to us are design or graphical students, Photoshop 2025 and Affinity were chosen, as Affinity can be viewed as an updated version of Photoshop.
After the pre-test, we explored how to create guidelines, which are then used for the second phase of testing, where these guidelines are used in a new method and then compared to the comparative test.

4.4.1 Pre-test and Comparative Test

The user test begins with an interview to evaluate the user's experience using the scale shown in Table 1.

Table 1: Experience scale used for tests.

1	2	3	4	5
No experience	Has used it at some point	Has used it for certain tasks	Has used it to a great extent	Uses it daily or professionally

The following questions were asked:

What is your experience with the program?

The purpose was to understand the user's understanding of their own ability and as a baseline to evaluate their experience.

What have you used the program for?

This was to understand what contexts their experience lies in, as Photoshop 2025 supports a wide range of use cases, it was vital to understand which part of the program the user had experience with.

Have you used any other similar types of programs? And in that case, which?

This was to see what their total level of experience in photo-editing is.

After this, we performed a think-aloud (Lewis, 1982) where the user was instructed to perform five tasks to the best of their ability, while also describing their thought process. The tasks focused on editing the picture seen in Figure 8.



Figure 8: The workspace of the test in Photoshop 2025. Image of a man downloaded from Pixabay.

The missions are as follows:

1. Remove the background of the picture (turn it transparent)
2. Replace the text with the color of the shirt.
3. Change the color of the shirt from green to blue.
4. Darken the image.
5. Create the appearance of a light source illuminating the subject from the right.

Following the tasks, an interview was conducted with the following questions:

How did you feel during the test?

This was to better understand the user's state of mind while they were performing the test.

What part of the test did you think was the toughest?

This was to gauge their understanding of the system and to see if several users thought similarly about the difficulty of the program.

Is there something you wish could be added to the program to help you with the missions?

This was to see what they perceived as the primary issue with the program.

The tests were set to take 15 minutes, due to time constraints from both the

researchers' and the users' perspectives.

4.4.2 Tests of Conceptual Methods

After finishing the comparative test, we needed to test our concept against it. Three tests were conducted with three different sets of guidelines:

- **Nielsen's Usability Heuristics**, as they are standardized, and it would be interesting to see how an experienced user would react to them.
- **User-Centered Nielsen's**, but changed to fit the language of the user, as we were worried that users would have a hard time understanding the design-focused language of those guidelines.
- **Our Guidelines** found from the pre-test, as we wanted to try our guidelines and how they weighed against Nielsen's.

They were tested on Affinity with the same reasoning above, but this time with only two participants for each guideline. The data gathered can then be compared to the comparative test to see how well it holds up against a normal User-Test. The user is tasked with exploring a prepared workspace in Affinity. While exploring each grouping of guidelines, the participant was asked to:

Nielsen's Usability Heuristics
Explaining whether the program follows or breaks each Nielsen guideline. The description of each one, as described by Nielsen in his heuristic workbook (Nielsen, 1994), was read out loud before placing the text in front of the participant.

User-Centered Nielen's
Answering questions asked about the system. These questions are based on the example questions described in the Heuristic Workbook (Nielsen, 1994).

Our Guidelines
Answering questions asked based on our guidelines, created from our exploration in the pre-test[9].

Each test was set to take around 15-20 minutes due to time constraints for both researchers and participants. A full list and description of each guideline and question used for each version can be found in Appendix 3.

[9] See Chapter 4.4.1.

4.5 REFINEMENT

For this phase, we analysed the results of the previous phase. We also created a booklet to summarize our method in a way that is easy to understand, shown in Figure 9.

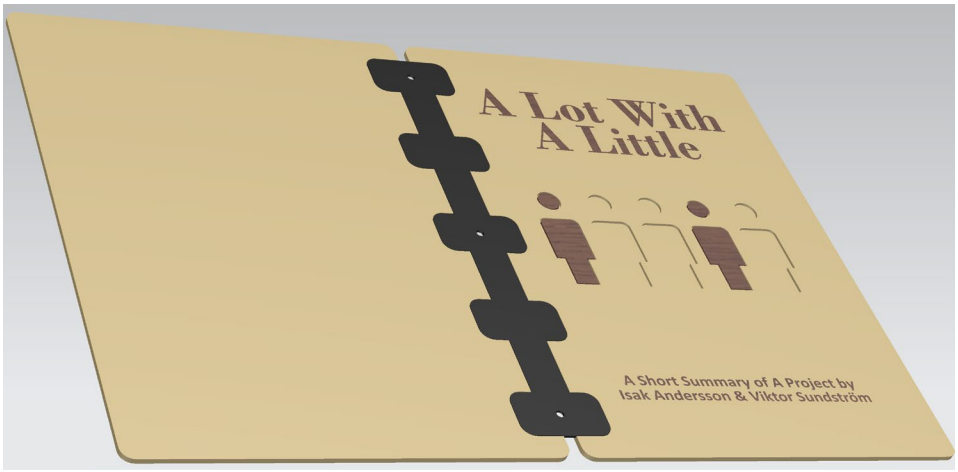


Figure 9: Booklet. Made in Siemens NX.

4.5.1 Analysis Method

For each problem found, we divided it according to which test it was found in, its severity, and its scope. Severity was based on the one created by Dumas and Redish (1999), but modified to better align with the structure of the test, shown in Table 2. Similarly, scope is also based on their work, with an extra point added as theirs originally included only the extremes of local and global problems, shown in Table 3. Problems were additionally grouped according to what type of problem they are, sorting them based on the menu to which they belonged (left, top, right), or outside of the menus. We also added a category for technical problems (such as bugs and glitches). The number of occurrences was also counted for each test.

Table 2: Severity.

1	2	3	4
Creates annoyance	Wastes some time	Creates a significant delay	Prevents the user from completing the task

Table 3: Scope.

1	2	3
Local Problem	Potentially global or part of a global problem	Global problem

In addition to sorting and counting the problems, we also compared the different tests according to total severity and scope across all identified issues. We compared the three methods, total severity, and scope, against the standard user test. This was calculated using the following functions:

$$Severity\ Function\ (EF) = \frac{Each\ Problem\ Multiplied\ by\ its\ Severity\ Rating}{Total\ Amount\ of\ Problems}$$

$$Scope\ Function\ (CF) = \frac{Each\ Problem\ Multiplied\ by\ its\ Scope\ Rating}{Total\ Amount\ of\ Problems}$$

$$Relative\ Severity\ (\%) = \frac{EF\ for\ a\ given\ Test}{EF\ for\ Comparative\ Test}$$

$$Relative\ Scope\ (\%) = 1 + CF\ for\ a\ given\ Test - CF\ for\ Comparative\ Test$$

Then the Relative Severity and Scope for the Comparative Test are set to 100%. This should then show the value of problems found by each method and show the quality of the problems found by the different methods, whether they bring up more important issues than a standard user test.

For the number of unique issues, there were a few problems that were only found by one of the different methods or user tests, with the assumption that the method with the highest number of unique issues found per user will be the method that should find the most issues during testing. This is calculated by:

$$Unique\ Issues\ per\ User = \frac{Unique\ Issues\ for\ Test}{Number\ of\ Users\ for\ that\ Test}$$

4.5.2 Booklet

As a summary of this thesis, we created a booklet. The intention is to spread the proposed ideas and methods beyond the scope of this thesis. The booklet consists of a summary of our theory, descriptions of our methods, and an example of use. It is created for the A5 format in InDesign 2024. The outside cover of the booklet was modeled in Siemens NX and manufactured using a Bambu Lab X1 Carbon 3D printer with AMS using their Wood PLA for the front and back, and their TPU 95 HF for the spine to provide increased flexibility. For inspiration, we read through Design: Process och Method (Wikberg-Nilsson, Ericson, & Törlind, 2017) and how it presents its methods. A full PDF of its contents can be found in Appendix 4.

4.6 METHOD DISCUSSION

Here, we discuss the consequences of the chosen methodology and possible alternative methodological choices.

4.6.1 Exploration

Literature reviews are standard for theses, but as ours is more theoretical in nature, this method became a central component of the project. For each phase, we spent time studying aspects of the information needed, such as improving evaluation approaches or conducting an in-depth study of our concepts.

A major challenge throughout the project was the vagueness of the path forward. This caused us to study a much broader area than initially anticipated. Over time, we realized that much of this material would not contribute to the final results. Because of this, there are parts that were studied that unfortunately did not lead to the results presented in this document. Together with this, there were a few theories that were found far too late to have an impact on the project. As an example, after we made our conclusions about magic numbers (which is to say, they are a sham), we found multiple articles that had come to the same conclusions. While we were happy that

many people agreed with us, it would have saved us a lot of time if we found those documents immediately. It also made us worried about any other information we might have missed. Which leads us to the next point:

There is always the possibility that relevant literature was missed due to the search not being sufficiently broad. As we just mentioned, we identified substantially more theoretical material than was required for the results of this project, which could indicate that our search was either too wide or not focused enough. In spite of this, we believe that the findings are robust and help contextualize the results. But even then, due to the theoretical nature of this thesis, there is a worry that we might have been too focused on theory. The problem with an overreliance on theory is that one eventually needs to prove all of this, which poses a methodological limitation for this project. Eventually, we had multiple theories that could be evaluated for this thesis, but due to time and budget constraints, we were unable to do so. Instead, we've opted to describe most of these paths in discussion, so that hopefully someone else can follow up with our questions.

The usage of AI is a widely criticized approach in the current day (Llerena-Izquierdo & Ayala-Carabajo, 2025). A major contributor to this view is infamous articles where AIs were used to write them entirely with no rewrites or proofreading, leading to a terrible article with dreaming sequences (Förster & Skop, 2025), a phenomenon commonly associated with generative AI. We understood the issues with this practice and instead aimed to use AI as another tool within our toolkit. As mentioned above, AI use as an advanced search engine makes it a powerful tool to find specific information without wading through pages of Google's recommendations or ads. In another sense, the problem isn't AI itself, but rather how it is used.

For our project, the AI model ChatGPT was used as an advanced search engine during the literature review phase. It was also used to lay the groundwork for the questions used during the different interviews. Due to the aforementioned dreaming aspect, we took the extra time to double-check each statement made by AI by reading the linked sources. In addition, we never assumed that the AI would be correct by its own statement and made sure to research any interesting statement.

4.6.2 Ideation

It should be mentioned that for each creative method we used, we changed something to better fit our goals. This is not uncommon among designers, but there is a natural worry about whether the changes remove the intended function of the method itself. This leads to a second line of reasoning, which is if that matters. If something is gained in the end, the specific approach may be less important.

For our methods, we chose ones that we haven't used before to see how they would play out. As novice practitioners, we take any moment we can to experiment with methods, and these seemed interesting enough. The Walt Disney method was chosen as an alternative to traditional brainstorming, as it structures creativity and encourages consideration of the problem from multiple angles. Analogies were used to generate additional ideas when initial ideation was insufficient. Finally, the Creative Matrix was applied to structure these ideas into coherent concepts. Even then, we needed to group these concepts afterwards to create two complete concepts.

While we believe our ideation phase was successful, it is possible that we could have

obtained more or better information if we had followed the methods exactly as written. A counterargument might be that creativity is not orderly and therefore should not be constrained to a specific manner of acting. Creativity is not a linear process and varies between individuals and groups. Creative methods function as guiding structures rather than rigid procedures, helping channel creative work toward productive outcomes without constraining it unnecessarily. The point is not that creative methods lack value, but rather that they should be changed to fit the user or group.

Despite this, the Creative Matrix was messy. It was possibly unnecessary to use both the analogies and the criteria at the same time, and would possibly have been enough to simply use the analogies and complete them with the criteria. Together with this, it should be noted that the criteria used during this part of the project would change later, raising the question of whether it was the correct move to use them in this part. Again, we gained a lot of concepts from this; it was still productive. Simply put, it would have been more proper to do it the other way.

4.6.3 Implementation

For this phase, we attempted to evaluate multiple aspects simultaneously. We performed a pre-test to test our historical theory, that old user data from a project can be used to create guidelines. The reason for combining these elements into a single test is our limitations, namely, time. We needed to get as much good data as possible out of one test, as we couldn't guarantee a second one. Due to this, there are a few other tests we were interested in doing, but could not fit into this test, and we unfortunately had to skip them.

Comparing a Heuristic Evaluation between Nielsen's Usability Heuristics and our guidelines.

This one was originally planned to be performed because it was considered an obvious comparison between these two methods, but unfortunately, the original user test took a week longer than expected, which meant we had to push this one back. Eventually, it was decided there was no time for it anymore.

Testing other guidelines with the method, such as Metaphors for Human Thinking.

We could have substituted Nielsen's for any number of guidelines, and there is an argument that we should have picked at least one more set of guidelines, as that would've given more information about how this method acts with different guidelines. As in, it would've shown if this method works best with our created guidelines or if it still would have worked with any set of guidelines.

Instead of using Photoshop and Affinity to simulate historical data, use an older version of Photoshop together with the newest.

While it would be more fitting to use an older version of Photoshop, it was decided that we simply did not have access to an old enough version of Photoshop for this to matter. The best would've been to use a Photoshop version from at least five years ago. This raises the question of why additional versions were not tested with multiple different types of versions to see the complete scope of this method.

Due to the aforementioned constraints, we understood that we could never get a complete picture of the things we are testing within this project. So instead of focusing on how these should work to function the best, we focused on evaluating whether

there was anything within these concepts that was worth continuing with. In other words, the focus was on whether the concepts were viable.

Outside of this, we also tried to use SUS for our historical test. We found no use for it, neither to create guidelines nor to help compare the different tests against each other. It could have been interesting to have it in the second part of the test to see how the perception of the interface changed between the different tests, but that was not important to see the viability of our method.

4.6.4 Refinement

We had largely defined what would be analysed from the previous phases, but it was natural that even then, we would discuss what more we could analyze from our data.

As an example, we knew that scope and severity were two factors we wanted to evaluate between problems. This was easy for severity, as we found a table to use. The only change came from what we found more intuitively understandable, meaning that a scale was preferred where 1 refers to low Severity and 4 refers to high severity, rather than the opposite. The problem came from scope, where we were unable to find any table or even a description of how to define whether something is or at what level of Scope. After much discussion, we settled on a 3-point system where 1 is a Local problem, and 3 is a Global one. The difficulty lay in defining whether a problem was truly global. Doing so would have required a deeper analysis of both the program and the identified issues. As this was beyond the scope of the project, a middle category was introduced to represent cases where globality was uncertain. If we had more time, we would've instead explored methods to define the global nature of problems.

Continuing from Scope and Severity, we also wanted a value that represented the amount found from each test compared to each other, which led to the functions described in Chapter 4.5. The problem is that they are a bit overly complex, making it hard to connect the results of them to what they're supposed to mean. While we do believe that the eventual graph showing it turned out well[10]. There remains a concern that people will overlook the results due to the complexity of the underlying functions that define it.

After analyzing all identified problems, we realized that an opportunity was missed to test the claim made by Kjeldskov, Skov, and Stage (2010), namely that experienced users identify more severe problems than inexperienced users. Although the number of occurrences was recorded for each problem, no distinction was made regarding which user identified each issue, as the analysis focused on which test revealed which problem. As a result, it was not possible to determine whether this claim applied to the present study, which could otherwise have provided insight into the accuracy of the user experience classifications or further supported the findings of Kjeldskov, Skov, and Stage (2010).

[10] Shown in Chapters 5.2 and 5.3.

5 Results

In this chapter, we present the results from the four phases: Exploration, Ideation, Implementation, and Refinement, following the Double Diamond framework.

5.1 RESULTS OF IDEATION

During this phase, multiple ideas were generated. These ideas were then analyzed and grouped into three categories according to what aspect of the project they focused on:

- Historical. Using past tests to further evaluate or structure a current test.
- Theoretical. Further defining the theory related to design tests.
- Visual. A tool to help visualize the project and problems.

We moved forward with the Historical and the Theoretical, as the Visual category fell outside the scope of the project.

5.1.1 Historical:

Some of these ideas were focused on using the history of the interface to gather specific types of data, either by maintaining an effective design revision history or by extracting meaningful information from previous changes. Building on these ideas, we examined potential uses of historical data. We first focused on studying statistical data because such data is typically well documented, making it easier to collect and analyze. We then examined whether historical design data could be used to identify which aspects could be considered important.

This developed into the idea that one could define the important aspects of the interface through old user tests. These aspects could then hopefully be used as a form of guideline for the system itself. These guidelines are intended to help designers focus on testing aspects of the system that are especially important and therefore find problems of greater severity and scope.

5.1.2 Theoretical

A few ideas focused on evaluating the test from a different perspective. Most of these were based on the concept of preparing users to optimize the information gathered to optimize the information gathered from them, or to deconstruct the project in such a manner that users aren't as important (or potentially avoided altogether).

The latter of these concepts had the limitation of already being well explored, resulting in well-known methods such as Nielsen's Heuristics and Cognitive Walkthrough. To explore an alternative approach, we examined whether these methods that avoid the user were used directly by users, falling into the first idea. This raised the question of how users would react to a method made for designers, which was a focus of this study. At the same time, we also wanted to see if and how these methods could be reformulated to be better understood by non-designers, as this could form a natural next step in evaluation and testing.

The rationale for this approach is that heuristics are grounded in intuition. Intuition is grounded in the evaluator's knowledge of the system. Because users draw on a different knowledge base than designers, their intuition is shaped by how they want

to use the design rather than how it is expected to be used. This intuition can be used to gain insight into users' perceptions of intended design use.

5.2 RESULTS OF IMPLEMENTATION

Here, we present the data gathered from the pre-test, the experiment with guideline creation, and subsequent tests. Variables mentioned throughout this chapter are shown in Table 4–6:

Table 4: Experience.

1	2	3	4	5
No experience	Has used it at some point	Has used it for certain tasks	Has used it to a great extent	Uses it daily or professionally

Table 5: Severity.

1	2	3	4
Creates annoyance	Wastes some time	Creates a significant delay	Prevents the user from completing the task

Table 6: Scope.

1	2	3
Local problem	Potentially global or part of the global problem	Global problem

5.2.1 Results of Pre-Test and Guideline Exploration

A standard user test was conducted with ten students at Luleå University of Technology. Participants had moderate experience with photo editing in Photoshop and a median experience rating of 3. A total of 22 problems were identified between them and categorised by severity and scope, shown in Appendix 3. From the test, 7 positive aspects were also noted. Discussion about bias that might have affected this test can be found in Chapter 6.3.3.

To identify guidelines, we focused on patterns in the issues we found that indicated important system characteristics. In total, we identified two methods for creating guidelines:

User-Focused Guidelines

From the user test, we observed a correlation between how the user wanted to use the system and how the system responded to the user's use. Guidelines were created to represent these behaviours and were then compared to issues found during the user test. From our test, we noted two distinct behaviours from users:

If the user didn't know what to do, they would take the time to explore the program, using the different learning tools available to try to understand how they should move forward. This gave us the guideline **Exploration and Learnability**, representing this tendency.

When experienced users knew what action to take, they sometimes selected an incorrect function or option. This could cause them to become stuck, as they did not

understand what had gone wrong. We also noted that inexperienced users had expectations for the system that simply weren't there. This behaviour was more difficult to define, but we eventually settled on **Expectations of image editing software** to represent the importance of creating something that both experienced and inexperienced users can understand the capabilities of.

Scope-Based Guidelines

Through our results, we managed to find three problems we defined as global, meaning that they occurred throughout the system (or they represent part of a larger problem with the whole system). Given this, it was therefore reasonable to assume that a guideline could be defined by this, a problem that exists everywhere. The three problems defined globally were as follows: *Difficulty in navigating to the desired function,* shortened to **Navigation**. *Need to learn the programme* was shortened to **Learning.** *Program-specific terms* were shortened to **Language.**

To better understand whether these are guidelines that fit the system itself, we asked two questions about each problem and possibility found during testing: Where the **Problem** is the given problem, where **Possibility** is the current possibility, and **Guideline** is the current guideline to which it is compared.

- *"The problem of [**Problem**] is a function of [**Guideline**]."*
- *"The possibility of [**Possibility**] is a function of [**Guideline**]."*

As an example of this, we found an issue with missing explanations of certain functions, which gave us the statement "The problem of missing explanations is a function of Exploration and Learnability". As the implication is that missing explanations prevent the user from exploring or learning the program, this statement was therefore considered valid. With another example, we found an issue with removing markings, giving us "The problem of removing markings is a function of Exploration and Learnability." While one could argue that this is a moment to learn, it is not directly connected to the idea of learning the user or allowing them to explore, instead being related to ambiguous presentation. We decided that this problem was not considered to align with the guidelines. In total, each guideline accounted for more than 50% of the identified issues found, indicating to us that they are valid. We moved forward with each guideline created from the pre-test and used them as Our Guidelines for the test.

5.2.2 Results of Test

In this chapter, we present the results of our test, together with the results of the comparative test.
For easier reading, each test has been shortened for the rest of this chapter according to Table 7.

Table 7: Names and descriptions of tests.

Colour	Signifier	Description
	Comp	The Comparative Test
	Niel	The test of Nielsen's Guidelines
	User	The test of the User-Centered Guidelines
	Ours	The test of our Guidelines

Comp identified the highest total number of issues, while **Ours** identified the fewest, as shown in Figure 10. When the number of issues is normalized per user, **Comp** identified the fewest issues (5.3 per user), whereas the other methods identified a similar number (7.5–9 issues per user).

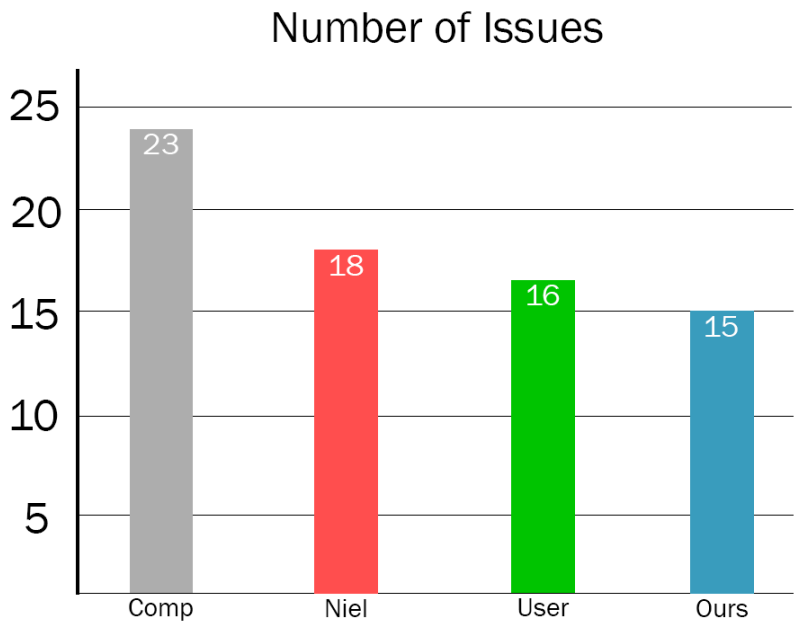


Figure 10: Number of problems found per test.

If we instead look at the number of severe problems found per test (Figure 11) and the scope of those problems (Figure 12), we will see that **Ours** found most of the severe problems from 3–4, and the highest number of problems with scope level 3. **Niel** found a substantial number among all severity levels (excluding severity level 3) and found most with Scopes 1 and 2. **User** identified a similar number of problems as **Niels** in the categories of severity and scope. These two methods differ most in problems with severity level 4 and scope level 2.

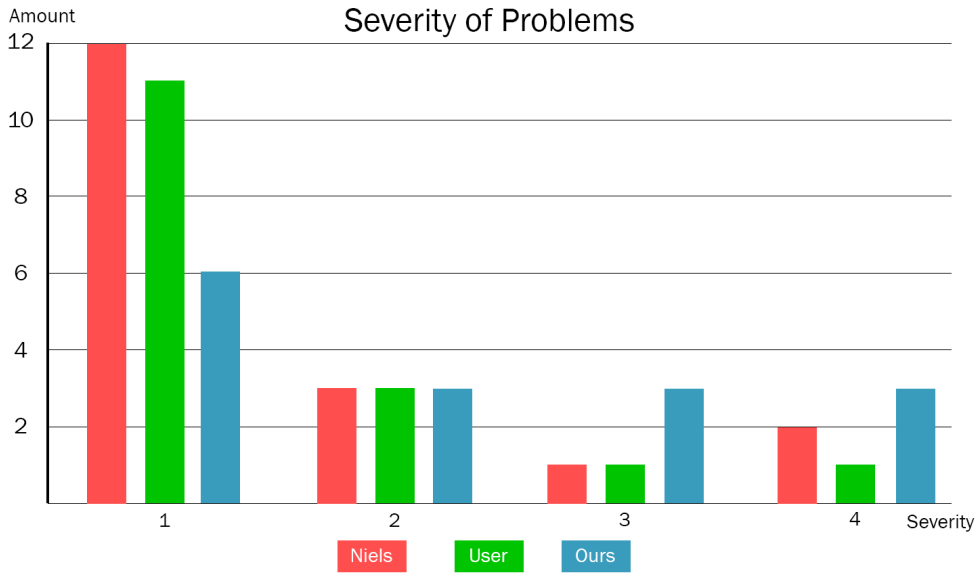


Figure 11: Severity of problems between different methods.

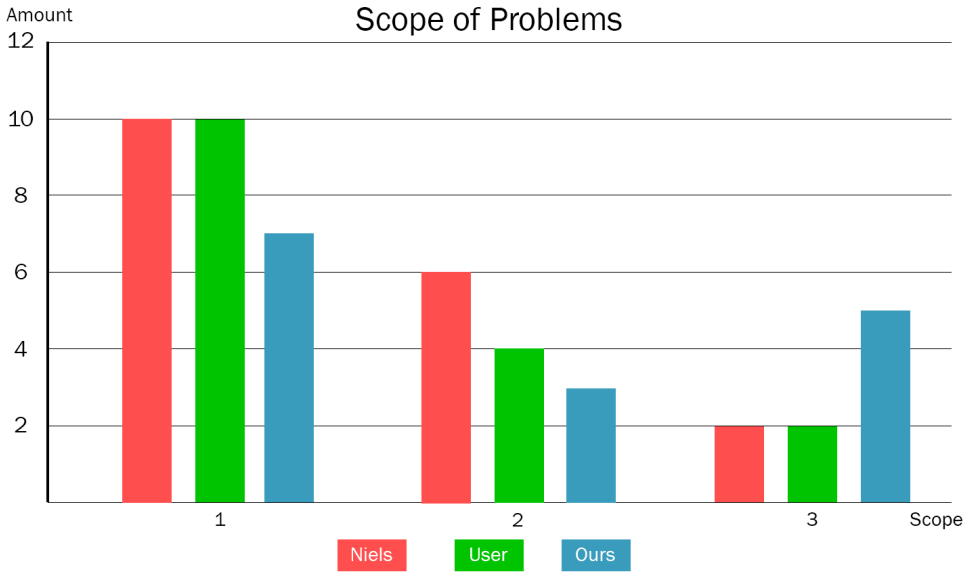


Figure 12: Scope of problems between the methods.

In addition, we can look at the number of possibilities found by each test, shown in Figure 13, where we see that **Niel**, **User**, and **Ours** found more than **Comp**. Together with this, **Ours** identified a substantially higher number than **Niels** and **User**:

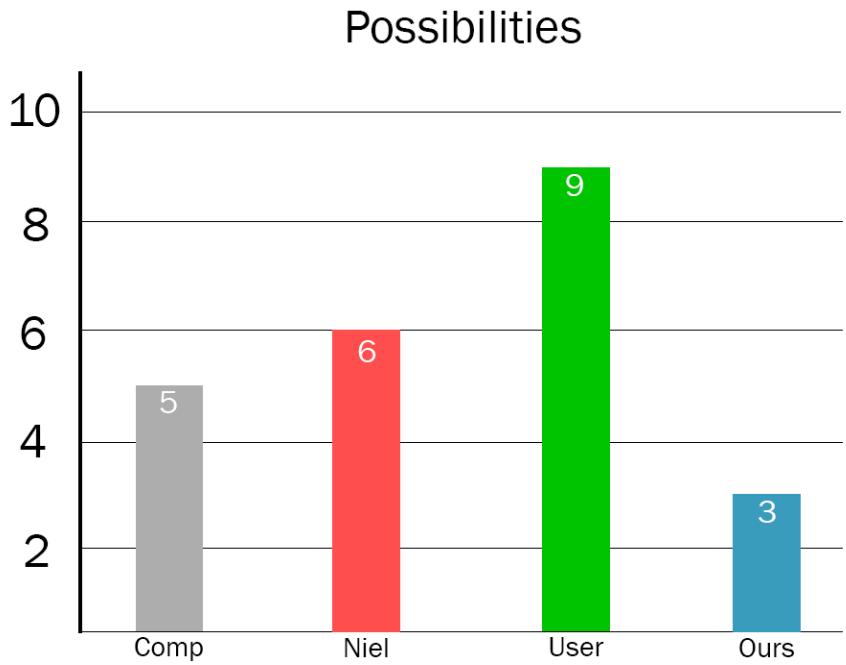


Figure 1313: Number of possibilities found in each test.

5.3 RESULTS OF REFINEMENT

Here, we present our analysis of the data described in Chapter 5.2. The names of the different tests are listed in Table 7, found in Chapter 5.2.2.

5.3.1 Analysis of Data

First, we examine the relative scope and severity of each problem identified according to which test they were found in, as shown in Figure 14. We see that **Ours** found the highest number in both severity and scope, while **Niels** and **User** found less than **Comp**.

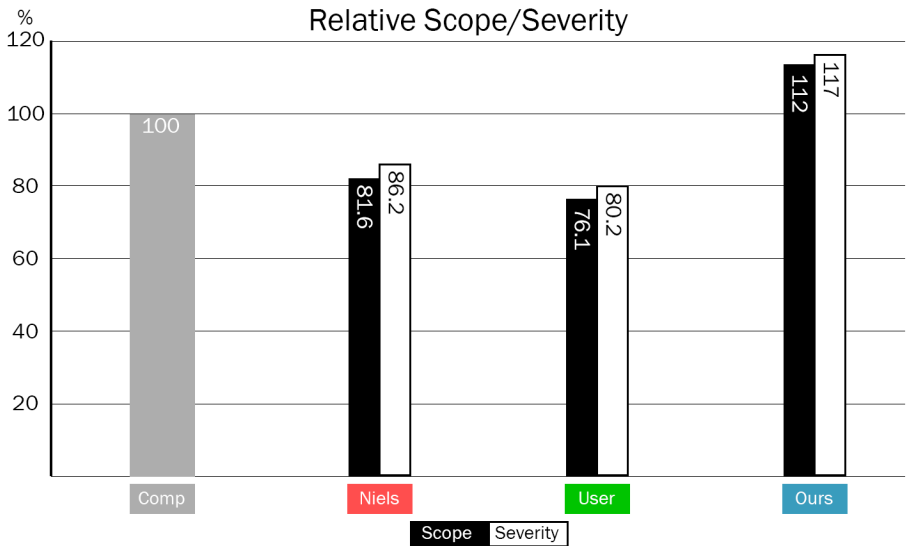


Figure 14: Difference in Scope and Severity between the different tests.

We can then look at the number of unique issues found in each test, as shown in Figure 15. We notice that **Comp** found the highest amount, but as **Comp** was performed by more users, it raises the question of how many unique issues are identified per user of each test. This is shown in Figure 16. Here, we observe that each test found a similar number, with **Ours** finding slightly more than the other methods.

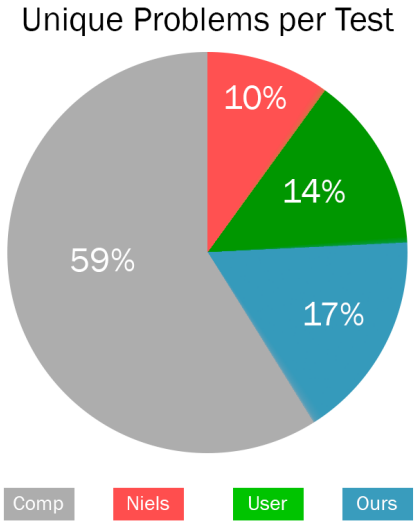


Figure 15: Unique Problems Per Test.

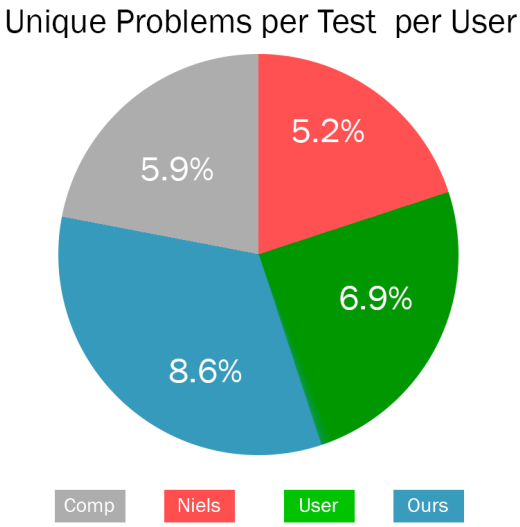


Figure 16: Unique Problems Per User Per Test

The types of problems identified by each test were also analysed. Problems were grouped according to menu location, absence from menus, or technical category[11], as shown in Figure 17. We notice that **Niels** found the highest number of problems in the left menu, while **Comp** found a substantially higher number of general problems with the program. Very few problems were identified in the top or right menus

[11] Not shown in the figure, as we don't know if the user found them due to the method or some other factor.

among all tests.

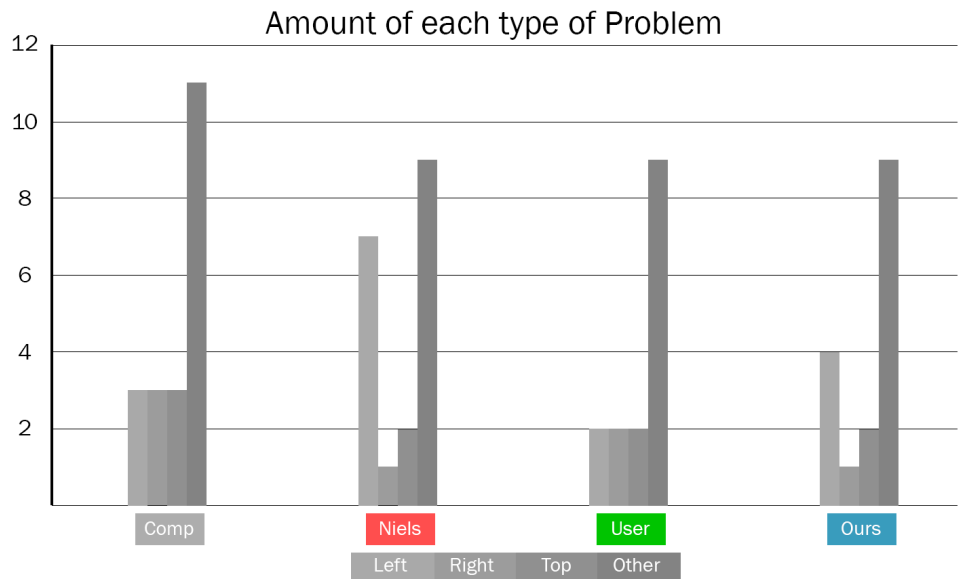


Figure 17: Types of problems and the amount for each test.

5.3.2 Analysis of Results

The following observations can be made:

The method for creating guidelines appears to function as intended.
These tests imply that our method of creating guidelines works. Further testing is required to determine how they perform, whether the method applies only to this test or to other applications, and how well.

Guidelines show promise when evaluating with fewer users.
Even outside Ours, using guidelines in a user test shows promise due to giving more unique issues per user than a standard user test. While the results show values very close to each other, which could indicate that other factors have influenced the results, the results from Ours show a substantially higher value. This may indicate that the approach is more effective within this testing framework.

Guidelines are understandable by users but require motivation for exploration.
Comp found a greater number of general issues, while the others found more problems in menus. It remains unclear whether this reflects user behaviour or the test structure itself. This is further discussed in Chapter 6.

5.4 FINAL RESULTS

In this chapter, are descriptions of our recommendations for how to use these methods, together with a short description of what the method is. For examples of how the methods can be applied, we urge you to read Chapter 4.4.1 or read our booklet in Appendix 4.

5.4.1 Specific Guidelines

Description:

This section presents two methods for creating project-specific guidelines. These guidelines can then be used in combination with established heuristics.

Prerequisites:

1. Gather user test data

This can either be from old user tests or data collected specifically for this method.

2. Evaluate each problem found:

Each problem should be ranked according to its severity and scope, as seen in Tables 5 and 6. Severity reflects the impact of a problem, including time loss and task failure, while Scope is whether the problem affects the whole system or only a small part. We recommend using the following scales:

Table 5: Severity

1	2	3	4
Creates annoyance	Wastes some time	Creates a significant delay	Prevents the user from completing the task

Table 6: Scope

1	2	3
Local problem	Potentially Global or part of the Global problem	Global problem

Method:

To create the guidelines, two applicable methods were identified:

Method 1: User-based guidelines

In this approach, guidelines are defined by how users intend to use the system. For problems with higher severity, evaluate what the user was trying to accomplish and how the system reacted. In other words, try to find where the mismatch between the user's expectations and the system's functionality or presentation.

Method 2: Scope-based Guidelines

In this approach, scope determines guideline definition. In this method, take each problem with the highest scope. These are your guidelines.

One could use either method, as designers may apply both approaches, depending on the availability of relevant data.

5.4.2 User-Centred Heuristic

Description:

This method is an alternative to Think-aloud, allowing users to perform a Heuristic Evaluation of the system themselves.

Prerequisites:

1. Choose guidelines:

To ease the evaluation for the user, choose what type of guidelines to base the evaluation on. Recommendations are Nielsen's Usability Heuristics and Specific Guidelines mentioned earlier.

2. Create guideline descriptions:

If no description already exists, create a clear description of each guideline. If possible, create cards or images that users can consult at any time. One could also create questions based on the guidelines, if they feel the need to.

Method:

3. Perform the user test

During the test, the user should have access to the interface so that they can explore it when needed.

For each guideline, present the description and allow the user to respond to it in any manner they choose.

Be prepared for questions from the user about the guideline; when questions arise, they should be answered by the moderator. But make sure that the guideline descriptions are consistent among users.

It is also important to allow the user sufficient time to think about the guidelines and how to connect them to the system. We also recommend using a usage scenario together with this to help the user become familiar with the system.

6 Discussion

Here, we discuss how our results relate to industry practice and affected stakeholders, in addition to how they compare to the theories discussed in this thesis.

6.1 POSITIONING THE RESULT

In Chapter 3.4, we explored a range of evaluation methods. These included approaches that evaluate systems without users, such as Heuristic Evaluation, Cognitive Walkthrough, and Action Analysis, as well as methods that involve users, such as User Testing, Think-Aloud, and the System Usability Scale. Compared to these, our methods fall into the group of those that have users, while borrowing elements of those that do not.

User-Centered Heuristics is based on Heuristic Evaluation (Nielsen & Molich, 1990), but as it is the user that performs it, it is more reminiscent of a Think-aloud (Lewis, 1982). The difference lies in the specific information gathered: With Think-aloud, the users' thoughts on why they perform a given action, together with comments about the design, are provided. User-Centered Heuristics draw on users' experience with similar designs, giving their opinion on them. In a way, this follows the ideal of co-design (Sanders & Stappers, 2008), where the users are experts themselves and should contribute to the design. Which redefines the type of expertise attributed to users. In this context, the user is an expert in using the product, while the designer is an expert in designing it. This is of particular importance, as it falls in line with Kjeldskov, Skov, and Stage's (2010) description of experienced users, who find more severe problems than those inexperienced. And in this context, the closer the experience is to the expected user, the more severe the problems the user should find. We should also mention Nielsen's (2012) comment about users:

"*Listening to what people say is misleading; you have to watch what they actually do.*"

With this method, we are going directly against this statement. While we understand that it is common for people to wish for things that they don't understand, the hope is that this method will help guide the user to the right mindspace, to give good suggestions about what they actually care about.

Specific Guidelines fall into the category of guidelines intended for use with Heuristic Evaluation, filling in another niche than those mentioned in this document. Usability Heuristics (Nielsen, 1994) are based on concepts that are considered important by the wider interface community, while Metaphors for Human Thinking (Hornbæck & Frøkjær, 2002) is based on psychological concepts. In chapter 2.2, we discuss the problems of Nielsen's Heuristics and their application, where we note that Nielsen has described them as "general" (Nielsen, 1994). In comparison, we found Metaphors for Human Thinking initially appeared novel, as it was previously unfamiliar. This issue emerged during a later interview with a usability researcher. In that discussion, it was noted that a method's complexity increases the risk of misuse. While Hornbæck and Frøkjær's (2002) guidelines are interesting, they are very hard to understand in comparison to Nielsen's Usability Heuristics. While one of our goals was to create easy-to-understand guidelines that can be used by anyone, we did not have time to

properly evaluate this aspect and can only wait and see whether they are actually easy or not. In either case, we hope that these different methods will help designers view problems from different perspectives.

The global method carries the problem of the globality of an issue being difficult to define (Dumas & Redish, 1999). We attempted to help define a method to do this while analyzing test user data, but the global guidelines might be a method to evaluate the globality of issues themselves. While we didn't have time to test this, the guidelines may become obsolete as more testing is done using them, as they are based on issues in the system. The eventual goal should be to remove these guidelines, rather than perpetuate them. The User-based guidelines show considerable promise but need to be better defined. The idea itself follows our description of User-Centered Heuristics, where the user is an expert (Sanders & Stappers, 2008), but we are still worried that they may be difficult to apply consistently. As we only had time to perform a single user test with this method, we were unable to perform any iterative improvement on it. This also made it harder to describe, as we are worried that the usage described is only applicable to the presented example.

We mentioned using either old or new user test data to create guidelines. While we were unable to properly use old user test data, instead opting for a substitute[12], we are still positive about its usage. At the same time, there are strong arguments that one should perform a new user test with the main goal of analyzing these aspects of the system (Kirakowski & Corbett, 1990). From this perspective, we contributed by developing a system for evaluating qualitative issues quantitatively. This approach is grounded in the descriptions provided by Tullis and Albert (2008). More directly, we've added a definition for levels of globality in problems. While we have discussed the problems with this, we hope that this will still help designers understand their usability issues better.

While we could not evaluate our methods through Hartson, Andre, and Williges (2001) method, we instead used a method more closely aligned with Siau and Rossi's (2011) approach. By comparing our methods to a normal user test through the same analysis method, we've used what they described as a "Theoretical and Conceptual Investigation." While evaluating the issues found from either test, we realized that an evaluative bias influenced how severity was assessed. As an example, both tests revealed that multiple users found the layer function in the program difficult to understand. The problem came from its severity: from the user heuristic, this issue prevented the user from completing the task (Severity 4), while from the user tasks, it only wasted their time (Severity 3). Intuitively, assigning the same severity would seem appropriate, but as we want to compare the value of each problem found, we also need to consider the differences between the same problems found. This difference probably comes from experience (Kjeldskov, Skov, & Stage, 2010), but due to the scope of this project, we were unable to explore this further.

6.2 RELEVANCE

The User Heuristics method is interesting from the user's perspective because it encourages analytical exploration of the system. Unlike a traditional user test, it does not ask users to simulate intended usage. This is relevant for the user as it puts more

[12] See chapter 4.4

value on the experiences they possess, and it elevates the role of users in usability testing environments. This could create an economic incentive for users to participate in testing. However, it may risk the integrity of evaluations if participants are not screened for alternative motives. This perspective aligns with co-design principles presented in Chapter 3.5.2. This could also have broader societal implications by assigning status to experienced users. A comparable phenomenon can be seen in influencers who receive sponsorships through product use.

The relevance of Specific Guidelines is more for the companies and the designer/researcher teams, as a parallel can be drawn to the reference point for a project's objectives. The guidelines are intended to help teams create awareness of what is important in the evaluation process. The interactive design community has described Nielsen's ten usability guidelines as obvious. In this context, our method may help companies develop their own explicit guidelines. This may contribute to sustainable organizational development. Internally defined guidelines can support cohesive communication and foster company growth.

6.3 REFLECTION

In this chapter, we reflect on our testing procedures, variable definitions, handling of bias, and evaluation of results.

6.3.1 Testing and Evaluating our Methods

One of the key factors affecting this study is the users in the test. As all participants were students or researchers, certain assumptions made from the results are specific to this user group. As an example, the guideline "Exploration and Learnability" may only exist due to the studious nature of the participants. Because the same participant group was used before and after testing, the results remain internally consistent. However, this assumes that the program is intended for this user group. This raises questions about whether the method applies to contexts with a broad user base. In other words, do our guidelines apply to image editors or students, or both? For this test, we assume that it only applies to students, but it may have some overlap with image editors. More testing regarding experience overlap is needed.

Our second set of guidelines, which consisted of an alternative description of Usability Heuristics[13], may have produced misleading results due to both the characteristics of the participants and our own role in the process. As we are also students, it is possible that the way we reworded these guidelines influenced their formulation, and that they were therefore adapted incorrectly. Nevertheless, the guidelines performed better than expected, likely because they were structured in a manner similar to the original heuristics. This outcome may also be explained by our similarity to the test participants, which in turn supports the argument that User-Centered guidelines can be effective when the evaluators share characteristics with the intended users.

Another approach proposed that instead of creating new guidelines, we should specify Nielsen's general guidelines to fit the program. To do this, we sorted each problem according to which heuristic it would fall under. The results were inconclusive due to limited data, and more testing would be required to see if this is possible.

Another problem we faced was validating these guidelines. For Scope-based

[13] Designated User-Centered Heuristics in Chapter 4.4.2.

guideline, the guidelines should work by virtue of being global problems. But the real problem came from User-Focused. We tried validating them by comparing them to the issues we had found, but we were unable to establish a clear relationship between the guidelines and the identified issues. We initially assumed that it would be fine since there was a majority overlap with the problems, but it was validated by a meeting with usability researchers, who responded positively to the guidelines. While this should imply that our method to validate them works, we would still like to perform more tests with this method to properly define it.

We realized early on that conducting a proper test using our methods would be challenging. In guided user tests, evaluators can more easily measure how long users take to move through an interface. Additionally, such tests allow evaluators to impose time limits on individual tasks, ensuring that the overall test duration remains within predefined limits. The idea for our test was that the user would guide themselves through the system in an attempt to either evaluate it according to the guidelines or to answer our questions. This made it really hard to create a maximum time limit, as we can't really know how long it would take for the user to understand each guideline/question, and then how long it would take for go through the interface. One possible solution would be to conduct a standard guided user test beforehand to familiarize participants with the system, followed by a response to the guidelines or questions. However, this approach risks introducing bias by directing attention toward specific aspects of the system. This reinforces the longstanding challenge of formulating questions without influencing user behaviour, a persistent issue in user testing.

Another aspect worth noting is the structure of our guidelines. They rely on questions rather than descriptive statements, which makes them closely resemble the think-aloud method. This applies to both the User-Centered and Specific guidelines[14]. This also makes them remarkably similar to Cognitive Walkthrough, especially the comment by Osvalder, Rose, and Karlsson (2019), where the user can partake in the method and become part of the evaluator team. The main reason was due to another worry about how one should structure a description in a way that guides the user into understanding it correctly, and also that we wanted to focus on seeing if the guideline itself actually works. This makes the User Heuristic approach more difficult to justify, but strengthens the case for Specific Guidelines. This does not imply that the User Heuristic method is ineffective; rather, that additional testing is required.

6.3.2 Defining Variables

Due to the nature of Scope, it is hard to evaluate whether a problem is global or not. We used a 3-point system: local, possibly global, or related to a global issue, and global. This became an issue during our evaluation process. The intent was to use scope ratings to create guidelines, which made this limitation more apparent. While we consider the idea itself to be sound, its foundation of identifying global problems is weak. Given the small scale of the test, it was fine to make a more assumptive decision, as there was less room for error. In larger-scale projects, this approach becomes insufficient. In such cases, global problems must be clearly defined. It can also be argued that one function of this method is to test whether a problem is global or not, in which case the challenge then lies in how such problems are identified.

[14] See Appendix 3.

Another interesting note is that we did not address the question of what the amount needed for a user test is. While we focused a lot on experience as an important part, we didn't explain how many are needed, given their experience. Although it may be possible to define a function that estimates the required number of users based on experience, we argue that the relationship is too complex to be adequately captured by a single function. That's why we prioritize experience and work with the participants available, and if additional participants are available, try to find those that fit best. In the end, it would be argued that the minimum can be defined by evaluators with experience in user testing, something that can be seen with Tullis and Albert (2008), who mention 5 users as a good minimum *from their own experience*. Whilst this number changes between projects, an experienced enough evaluator would possibly know when enough data has been gathered in any scenario. This leads to a new question: Is there some effective way to pass this information on?

6.3.3 Biases

Due to our constraints (that include area, time, and budget), the majority of the users we gathered for the tests were students. This was a deliberate choice, as students are expected to use such software. Adobe products are widely used in universities, and testing on university grounds reflects the program's typical usage context. The main limitation is that most participants were Industrial Design Engineering students. As such, they had prior experience with usability testing, which challenges the assumption that participants only had experience with image-editing software. One could argue that this limitation might be mitigated by assuming prior training from the user in this context. However, such an assumption is not appropriate for this type of test, as we wanted to evaluate with as few experiences as possible. At the same time, the primary aim was to test whether the theory could function in practice. The results still indicate something about the usability, but instead indicate whether user preparation is necessary to be properly evaluated or not.

As we are students ourselves, this also raises questions about our relationship to the testing process and results. Before testing, we held preconceived opinions about the system. These may have influenced both problem identification and severity or scope assessments. We tried alleviating this by clearly defining the different levels of Severity and Scope. This may have led to over- or underestimation of problems. Addressing this would require a larger and more diverse group of evaluators. Another limitation is that there might have been a problem with how we evaluated the user's experience. We assessed user experience through self-report and observation. A discrepancy between these measures became apparent. In retrospect, a more detailed method was required a more in-depth method to evaluate it. However, time constraints prevented proper implementation.

As an interesting note, the menu where the most problems were found was in the tool's menu. This could be because it is inviting in appearance, which is to say that most users with any amount of experience in image-editing software usually look there first. An additional observation, this was the only menu that had a notable number of people looking into it. The number of issues identified in this menu varied significantly between tests. In one test, only two problems were identified, whereas Nielsen's test identified eight. This raises the question of whether users are more inclined to explore the left menu without explicit instruction. This may be because the left menu relies primarily on icons. In contrast, the other menus rely mainly on text, which may appear less approachable.

6.4 CONCLUSIONS

In this chapter, we present what we have found in the context of our objective, aims, and research questions.

6.4.1 Project objective and aims

Our objective is:

To evaluate existing usability methodologies and to develop a method that helps designers and researchers gather information with a low sample size.

In this thesis, we have explored and evaluated different usability methods according to their capabilities. Although we explored how to evaluate these methods, we chose not to pursue this further. Instead, we focused on developing our own methods. This has led to the exploration being wide, yet also not that deep. This indicates that further work remains in this area. We have collected and described multiple user-centered methods; the remaining task is to systematically compare these methods

At the same time, we went beyond the initial scope and created two methods that can be used when you have a low sample size. Both methods require further exploration to clarify their applications. Nevertheless, the results indicate potential effectiveness. Specific Guidelines need further exploration into the creation of the guidelines themselves and how to evaluate the guidelines. User-Centred Heuristics need to be tested with more types of guidelines and further explored together with scenarios. Despite this, the user test data indicate that the methods function well together. This result is both encouraging and concerning. While it is good that this might work, there is a possibility that the methods are only effective when used together. Even then, the exploration of these methods has opened opportunities for others to explore similar approaches.

Given the breadth of existing methods, similar approaches may have been proposed previously. However, it is not feasible to identify all prior work. If similar methods already exist, our contribution may still help increase their visibility and relevance.

Aim: *To help designers and researchers perform usability evaluations when there are a few users available for testing.*

Through our exploration of multiple methods, we identified which user characteristics yield the most valuable information and what factors matter most when selecting participants for usability tests. In summary, the expected user profile should be clearly defined. Participants with experience similar to the user profile should then be selected. The importance here lies not in the amount, but rather in the experience of the users themselves. In cases where the interface is expected to have highly trained users, as few as two participants may be sufficient, given that they are experienced enough. It should be noted that there are additional factors that are likely to influence how many problems a user can find. Within this document, we've also explored bias and how that affects experience, with the conclusion that understanding what bias exists can prevent it from hurting your user tests too much or help you in gathering better data. There is more that can be explored in this area, such as how much bias can affect user tests and if it is possible to debias a user[15].

[15] Something we discussed partly in Chapter 3.4.3.

Aim: *For one of our methods to widen the method landscape, for when organizations and companies are worried about not having enough resources to evaluate properly.*

Due to project constraints, no budgetary analysis was conducted. This limits our ability to predict organizational applicability. Even then, we believe that the theory can help companies choose better users, which in turn should allow them to use fewer users and therefore create a lighter budget. While being in contact with a group of designers and usability researchers, they seemed positive about the results of this document. This suggests that the work may have had some impact.

6.4.2 How can usability researchers interpret and prioritize usability problems while working with fewer than five participants?

In general, the problem does not lie with the number of users, but with the type of users. Experienced users find fewer problems than a general user, but these problems are more severe than those of a general user (Kjeldskov, Skov, & Stage, 2010). The implication is that a smaller number of users than initially expected is required, provided that they are experienced in the context for which the product is intended. Additionally, we recommend evaluating problems according to their severity, using Table 5:

Table 8: Severity.

1	2	3	4
Creates annoyance	Wastes some time	Creates a significant delay	Prevents the user from completing the task

Problems should be addressed according to their severity. While we do recommend evaluating a problem through its scope, we can only say that this requires more extensive analysis. We will still recommend Table 6, which we used for our tests:

Table 9: Scope.

1	2	3
Local problem	Potentially global or part of the global problem	Global problem

Problems with a scope rating of 2 or 3 should be investigated more closely. Problems rated as scope 1 generally require no additional measures, aside from their severity score.

6.4.3 What aspects of a digital interface can be evaluated without users?

Numerous methods allow a designer to evaluate a design before a user touches it, but they all carry the same general problem: Without user input, a designer cannot reliably know what a user will think about a system. Nor can they fully predict how it will be used. Some methods attempt to mitigate this limitation, such as Action Analysis and Cognitive Walkthrough. However, these methods can never fully replace user involvement due to differences in designer experience. In other words, the designer's understanding of the product is not guaranteed to overlap with the user's understanding (Norman, 2002).

However, this does not mean that these methods are without value. Most of these methods can help identify general or obvious issues with a system, without defining

the issues according to their scope and severity (Hertzum, 2006). These problems may not need to be evaluated according to their budgetary cost, as they need to be addressed regardless. To evaluate its importance, a user test can be performed to evaluate it. Even a single test can be sufficient to clarify this. If you are interested in more in-depth descriptions and analyses of different methods, these descriptions can be found in Chapter 3.4.

6.4.4 How do other research fields mitigate the limitations of studies with fewer than five participants, and how could these approaches be applied to usability evaluation?

The question mainly relates to the availability of participants and the context in which they use the system. Psychology provides another relevant example. In this field, the number of participants is often debated, as some studies focus on a single individual. Dukes (1965) discusses several single-subject studies in psychology. These cases demonstrate that such approaches can still produce valuable insights. Clinical medicine is another area where a low number of users is a possibility. This is particularly evident in general medical examinations, where a single patient is assessed to identify symptoms and determine a diagnosis.

In both cases, the key argument is the uniqueness of the individual. The focus is on addressing the needs of a single person. When applied to design, this argument refers to situations in which a product is created for a specific individual. A clear example is prosthetics, which are designed for a single user, even though the general concept of the product may vary.

Another argument concerns generality. In this case, a single user may represent a broader user group. In design, this may apply to products that require specialized training. In such cases, users tend to share similar experience levels. As training requirements increase, this assumption becomes more valid.

6.4.5 How can one avoid bias affecting the results?

Unfortunately, bias cannot be fully avoided when evaluating systems, whether it originates from users or designers. What can be addressed instead is an understanding of which biases may affect the testing environment, and it would allow one to better evaluate problems. We followed a list created by Tullis and Albert (2008), which lists several sources of bias[16] that could affect the area.

At the same time, one could also account for bias strategically. Knowledge and experience inherently introduce bias. As a result, certain biases can be anticipated among users with specific experience levels. As an example, during our testing of image-editing software, we realized that the more experienced the users were, the more they expected the program to act a certain way. If the goal were to improve the program, these expectations would warrant closer examination. This would involve identifying which behaviours are expected of software of this type, to create something that those users would find easy to use.

With only a few experienced users, there is a risk that bias could disproportionately influence results. In such cases, the design might become overly tailored to the specific test participants. This aligns with the findings of Dukes (1965), who suggests that an experienced user may be representative of a broader user group when high levels of

[16] See chapter 3.4.3 for a full description.

expertise are required. In such cases, bias may have a smaller impact than expected. This is because both the system and its users require a high level of expertise.

6.5 FURTHER STUDY

Due to the limitations of our research, we were unable to explore every aspect of the project. Feedback from several researchers, who suggested that this work could form the basis of a doctoral thesis, further highlighted that the project scope was greater than initially anticipated. The following points could not be addressed within this project and could be expanded upon.

6.5.1 General

- Explore Nielsen's binomial function (1993) and assess whether it can be extended. Identify which additional variables would be required and whether they can be described mathematically. If this is not feasible, investigate alternative methods for estimating the required number of users.
- Explore Molich et al's (2004) statement that more user tests should give more information than more test users. How could one structure multiple user tests with a small group of users? What problems would come from using this method? And how could one alleviate them?

6.5.2 Specific Guidelines

- Explore how guidelines can be created from data analysis.
- Explore how a guideline can be defined as valid for a project.
- Explore how user domain knowledge affects the creation of guidelines.
- Explore how other applications are there for guidelines? Not only Specific, but also Nielsen's or others.
- Explore how Specific Guidelines work with more general users, rather than those of specific experience.
- Explore methods to evaluate the Scope of a problem.

6.5.3 User-Centered Heuristics

- How to best structure a scenario to allow the user to still consider the guidelines?
- How well do other guidelines perform when used with this method? Does the method function best when applied specifically with Specific Guidelines?
- Create a method to evaluate the user's level of experience.

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1. Career Journey
   - *How long have you been working in the UX field? How long have you been a designer? How long have you been a designer for this company?*
     1. *What have you designed while working for this company?*
     2. *What are you currently working on?*
        (If you are allowed to reveal that, of course.)
2. Pipeline & Process
   - *"Can you describe your UI/UX pipeline from research to delivery, especially for high-stakes products? How does it differ from more traditional consumer-focused pipelines?"*
   - *How does your general design pipeline look? / What pipeline did you use for your project?*
3. Trained Users vs General Users
   - *How much do you work with trained users and untrained groups in your projects?*
   - *While conducting user tests, have you used both trained and untrained users? Have you ever done it during the same project?* ← Skip if it has already come up.
     1. *What would you say is a key difference between evaluating these two groups?*
        What are the factors that apply to one of them but not the other?
4. Cognitive Demands
   - *"When designing for users under stress or high workload, what are the most beneficial methods in evaluating the interface?"*
5. Small Test Groups
   - *In your experience, when is it valid to work with smaller-than-standard test groups, and how do you compensate for limited participant numbers?*
   - *We recently read the article by Nielsen about why 5 is considered the minimum number of test-users. Is this your experience, or do you generally have a different amount?*
     1. *More specifically, when do you feel it is necessary to use more than 5, and when do you think that less than 5 is enough to evaluate?*
     1. *Even more specifically, what factors are it that affect your decision-making when it comes to the number of test-users required to evaluate a certain design aspect?*
   - *While designing for small test groups, what would you say are the most important aspects to keep in mind?*
8. Methods with Few Users
   - *Have you experienced situations where user access was limited or unavailable? What methods did you use to evaluate your designs in those cases, and why?*
   - While creating a design, have you ever run into the trouble of low to no test-users?
     How did you evaluate the design then?
9. Limits of Evaluation
   - *Are there aspects of interface evaluation that, in your opinion, simply cannot be done without access to a sufficient number of users? What are those, and how do you mitigate the risk?*
   - *Are there any aspects of a design you think would be harder, or even impossible to evaluate without enough users?*
     1. [If Yes]
        Are there any aspects you would say would not be possible to evaluate with fewer than five users?
     2. [If No]
        Are there any aspects you would say require more than five users to evaluate?

Heuristics: See Nielsen (1994).
User-Centered, Example Questions:

Visibility of System Status

Does Affinity explain what it is doing at any given moment? How?
Does Affinity explain what happens when you use it? How?
Do you feel that Affinity is responsive? What do you see Affinity doing that is responsive?

Match Between System and the Real World

Read through the menu on the left—read through the top menu—read through the menu on the right. Do you understand their function based on their names?
Are these accurate representations of the various functions in the menus? Is there anything that isn't?

User Control and Freedom

Once you have created an object, can you go back and change how the object was created?
When you start a process, can you see how you would cancel it? Can you do it easily?
Are there undo and redo functions?

Consistency and Standards

Does Affinity's layout mimic the layout of other photo editing programs? Are the menus where they usually are? Are the icons similar? Are the names the same?
Are the visual elements of the interface consistent throughout the program?

Error Prevention

How difficult is it to do something you didn't mean to do?
Are there features in Affinity that you can't undo? Does Affinity warn you about this?

Recognition Rather Than Recall

Does Affinity keep important information visible so you don't have to memorize it? What about information that is important to others?
Is help available if you get lost?

Flexibility and Efficiency of Use

Are there shortcuts to different functions?
Can you customize Affinity's menus and tools for individual purposes?

Aesthetic and Minimalist Design

When does Affinity display more information than necessary?
Is there anything you would call unnecessary on the screen?

Help Users Recognize, Diagnose, and Recover from Errors

Are error messages clearly visible?
Does Affinity provide suggestions on how to resolve error messages?

Help and Documentation

Is it easy to find what you are looking for in the help documentation?
Does Affinity provide help when you need it?

Specific Guidelines, Example Questions:

Exploration and Learning

Do you feel anxious about exploring the program?
No: What does Affinity do to make you feel secure?
Yes: Why does Affinity make you feel anxious?
Do you feel that you can learn from the resources Affinity provides?
Do you think others would do the same?
Can you try to find a specific feature?
Did you feel that the program guided you? Do you understand how to get there?

Expectations of image editing software

What does Affinity expect you to know about image editing?
What makes you feel that way?
Do the icons and names resemble other photo editing software?
Do you understand the different menus?
(If they are not looking to the right, nudge them there)

Navigation

Does the grouping of items seem logical?
How does Affinity allow you to navigate to a specific function if you don't know where it is?
How has Affinity grouped its menus, and how do you think others understand how they are grouped?
Are frequently used tools easier to access than tools that are used less often?

Learnability

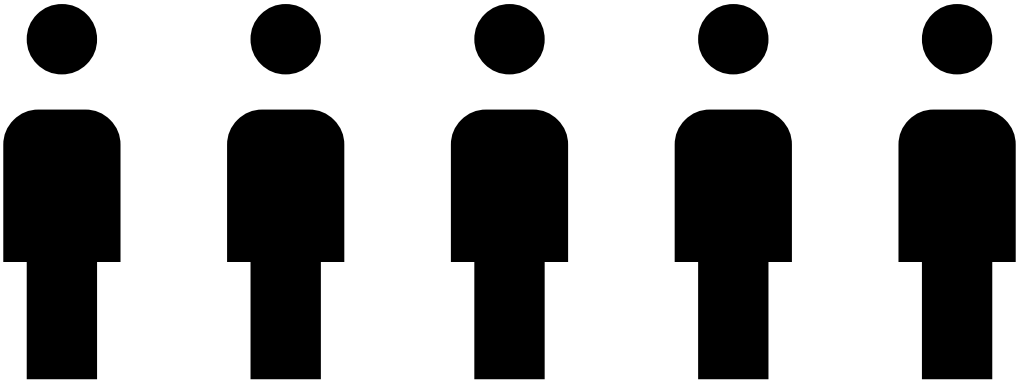
[Skipping due to similarity to exploration and learning]

Language

[Uses the same questions as User-Centered's Match between System the Real world]

Test	Problem	Viktning	Scope	Test
User-Heuristic	"Error"-ruta frånkopplat från område	1	1	1
	Tar längre tid att få upp menyer då man har redan valt verktyg.	1	1	1
	Otydligt Namn/Beskrivet	1	2	1
	Undangömda Genvägar	1	2	1
	"No Camera Data" visar inte viktig information	1	2	1
	Oresponsivt Verktyg	2	2	1
	Svårt att ta bort markering	4	1	1,3
	Svårt att förstå histogram	1	1	2
	Ingen snabb metod för att byta plats på menyer	1	1	2
	Otydlighet i vad undo reagerar på	1	2	2
	Fattas förklaringar i delar av programmet	1	2	2
	Svårt att snabbt kunna stänga av alignment	2	1	2
	Tar tid att hitta "ingen färg" iconen	2	1	3
	Ingen Guide för Navigation	2	3	3
	Förväntar sig insikt i termer och språk	3	3	3
	Nybörjare Tillgodoses Ej	3	3	3
	AI-sökning fungerar Ej	4	1	3
	Flyttat runt på funktioner	4	2	3
	Video fattas	1	1	1,2
	Otydlig Sökfunktion	2	1	1,2
	Otydlighet med Lager	4	1	1,2
	Otydligt då det har blivit fel.	1	1	1,2,3
	Meny näst-högst-upp för stor	1	1	1,2,3
	Sökfunktion väljer inte verktyg Automatiskt	1	1	1,2,3
	Pop-up rutor försvinner	1	2	1,2,3
	Annorlunda Ikoner/Namn	1	3	1,2,3
	Ingen uppmaning till att lära sig via programmet	1	3	1,2,3
	Svårt att urskilja felmeddelanden	2	2	1,2,3
	Otillgängliga verktyg visas som tillgängliga	3	1	1,2,3
Comparative	Ovalbart Verktyg Placerad Först	4	1	1
	Selection låses till lager	4	1	1
	Otydligt hur verktyg fungerar	1	1	1
	Kunde inte förstå vart man flyttade i lightnings func.	1	1	1
	Ingen "Direct Search"	1	1	1
	Flood-select baserad på prestanda	2	2	1
	AI inte ordentligt integrerat i program.	2	2	1
	Otydligt var man ändrar pensel-storlek	2	2	1
	Tar lång tid för informationstext att visa sig	1	2	1
	Ingen Hjälpinformation i Övre Meny	1	2	1
	Kunde inte bara få upp verktyg i hjälp (separat sök)	1	1	2
	Otydligt var man gör invert-selection	3	1	2
	Undangömd Error Pop-Up	1	2	2
	Osäker färgfunktion	1	2	2
	Error-popup försvinner för snabbt	1	2	2
	Prestandabaserade Funktioner	1	3	2
	Gradient heter Fill istället	1	1	3
	Svårt att skanna för sådana med dyslexi	2	3	3
	Annorlunda Ikoner/Namn än Photoshop	1	3	3
	Otydligt lagerskapande (omedvetande)	2	2	5
	Otydligt lagerskapande (medventande)	3	1	6
	Annorlunda funktioner från andra bildredigeringsprogram	3	3	7
	Svårt att avmarkera	4	1	8

A Lot With A Little



A short summary of a project by:
Isak Andersson and Viktor Sundström

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About this Booklet

This Booklet summarizes A Lot with A Little - Evaluating Usability Methods for Developing Two New Methods for Small Test Groups, the Thesis Project of Isak and Viktor. In here, you will gain a better understanding of how to gather the right Users for your user test and will learn of two new methods to evaluate them.

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Choosing the Right User

How many are too few?

There is a certain expectation in user testing that more users will bring more information. And it makes sense, right? More people giving their opinions about a design. But generally, this does not hold true. When performing user tests, we aren't necessarily looking for more information, but better.

And what would be better information? In the context of user tests, it would be very severe problems, problems that waste a lot of time or prevent the user from using the design correctly. But even then, how do we know how many people are needed?

For quantitative evaluation, you want as many users as you can get. As you are looking at it from a statistical point of view, more users help generalize the data, which gives more concrete data.

> *What matters more than amount is the quality of the users themselves.*

But for qualitative evaluation, what matters more than the amount is the quality of the users themselves. Quality in this case refers to how well the user fits into the expected user demographic. But in certain cases, the user demographic is so wide that most people would fall under it, such as for a public website meant for everyone. In this case, a lot of people are still needed due to the wide range of expected users.

So, to summarize:

- Instead of looking for a certain amount, look for the right type of user.
- Quantitative data still requires a good number of people, while Qualitative data cares more about the importance of the right type of user.

What makes a great user?

A great user varies depending on the project. One could base it on the expected user to get a good image of a great user, but as in the example mentioned earlier, this isn't always viable.

As an example, take a car. What type of user would drive a car? Well, they would have a driver's license, so they can use the car. They would also preferably have had some time driving those cars, enough that they are used to using one. This leaves out a lot of people, but it showcases what is expected of the users we want. Now we can get into the nitty-gritty, what type of car do we want to make? If it's a race car, we would need race car drivers to test it. If a taxi cab, we would need a taxi driver to test it, so on and so forth.

> *While we are experts in design, they're experts in using that one product.*

If we were to put all of this into a single word, I'd choose experience. The experience they have of using the product we want to test. To put it into other words, while we are experts in design, they are experts in using that one product. And we need to convert that knowledge into something usable from a design perspective.

Of course, we don't always want only experts. Sometimes we want a wider net, as with the example earlier with the website. Generally, anyone should be able to use a website, and the group of anyone carries a lot of different types of experience. Some of them might be IT-experts, others might be someone's tech-illiterate grandmother.

In either case, one should define the expected user and try to find someone who defines it clearly. If you want a product for experts, find experts! If you want a product for anyone, find anyone!

Specific Guidelines

What is it?

Throughout design, there have come certain guidelines that can be used to better focus on aspects that are important of a design. The most famous example would be Nielsen's Usability Heuristics, a group of 10 guidelines that apply to any interface.

The problem with most of the guidelines is that they are general. Which is to say, they come with certain aspects of the design itself. This causes certain of these guidelines to not be applicable in all cases.

To solve this, one could create a set of Specific Guidelines for a system, instead of using a group of general ones. This would guarantee that they are useful for the project.

When should I use it?

During a larger project where standardized guidelines have a hard time fitting, or you simply need more.
Also if you either have test data or can gather it. It can be from normal user tests, but preferably data from at least one test with 10 users.

Time = 1 - 7 days, depending on the size of the project.

How?

1. *Gather User Test Data*

   This can either be from old user tests or ones you gather for this method.

2. *Evaluate each problem found.*

   Each problem should be ranked according to its Severity and Scope. Severity is how much time is lost, while Scope is whether the problem affects the whole system or only a small part. We recommend using the following scales:

Severity	1	2	3	4
Description	Creates annoyance	Wastes some time	Creates a significant delay	Prevents task completion

Scope	1	2	3
Description	Local	Maybe Global or part of Global	Global

3. *Define Guidelines according to one of two methods*

   Scope Based

   Each problem with the highest scope is a Guideline.

   User Based

   Evaluate the wants and needs of the user. Why did the problems appear? What were they trying to accomplish? These wants and needs become guidelines.

4. *Use the Guidelines*

   What you want to do with the guidelines is up to you, but examples include:

   - Inspiration for a user test
   - Heuristic evaluation
   - Simply designing around these guidelines

Example of Specific Guidelines

As we had no previous user tests to work from, we created a new one. Here, we performed a basic user test on an image editing software, where the user performed a Think-aloud while working on five tasks. After all that, a brief interview was conducted to gather feedback about their experience using the program.

In total, we found 22 issues. We sorted these issues according to their severity and scope and found the following:

Scope Based

Out of the 22 issues, three were global. These problems were as follows:

Issue	Description
"Difficulty in navigating to the desired function."	Everyone had problems finding what they wanted.
"Need to learn the program."	Multiple users commented that they needed to learn the program to use it correctly.
"Program-specific terms."	Words used in the program are difficult to understand if you are not knowledgeable in the subject.

For each of these problems, we gave it a title to make it easier to understand what the guideline was. Each problem became in order:

- ***Navigation***
- ***Learning***
- ***Language***

User Based

Throughout the test and while looking at the issues found, we tried to understand how the user wanted to use the system and how the system reacted to this. Looking at it through this lens, we saw:

If the user didn't know what to do, they'd take the time to explore the program, using the different learning tools available to try to understand how they should complete the task. This created the guideline ***Exploration and Learnability***, representing this tendency.

If the user knew what to do, they would still mistake how a function worked. This would sometimes cause the user to become stuck, as they had trouble understanding what had gone wrong. Together with this, a lot of users had expectations of the software that simply weren't there. We assumed it had to do with the assumptions of the system, giving the name to ***Expectations of Image Editing Software.***

In summary, we gained a total of five new guidelines from this method, that are as follows:

- ***Navigation***
- ***Learning***
- ***Language***
- ***Exploration and Learnability***
- ***Expectations of Image Editing Software***

User Heuristics

What is it?

Heuristics is a method wherein a person examines an interface and decides what parts of it is good or bad. To help with these decisions, they might use a set of guidelines that define what parts should be looked at, such as the ones mentioned earlier.

The reason this works is that as designers, we have an understanding of how these things are supposed to function. Not maybe from a mechanical perspective, but rather a human perspective. We understand what a human might want from it.
But at the same time, we don't know exactly what a user might want from it, a human who either wants or has to use the product.

A user has a completely different set of knowledge that they apply to the product, which is equally as important as the one the designers have. This method attempts to utilize this knowledge to further our understanding of the interaction between the user and the product.

When should I use it?

When you are working with a complex product, and you don't have access to many users, but the users you have access to are experts.

Time = 20 - 60 minutes per guideline.

How?

1. *Prerequisites*

1.*A Choose guidelines*

To ease the evaluation for the user, choose a pair of guidelines to base it on. Recommendations are Nielsen's Usability Heuristics and Specific Guidelines mentioned earlier.

1.B *Create guideline descriptions*

If they don't have it already, create a clear description for each guideline. If possible, create cards or images where the user can read the description at any point.

2. *Perform the User Test*

For this part, the user should have access to the interface so that they can explore it when needed.

For each guideline, present the description and allow the user to respond to it in any manner they like. Be prepared that they might have questions about the guidelines, which should be answered. But make sure that the description of the guideline is consistent among users. Remember to also give the user time to think about the guidelines and how to connect them to the system.

We also recommend using some kind of scenario together with this to help the user get into the system itself.

Example of User Heuristic

For our test, we tested an image editing program with Nielsen's Usability Heuristics and our Specific Guidelines, shown earlier. For the Usability Guidelines, we opted to use the descriptions found in the Heuristic Evaluation Workbook[1]. For the Specific Guidelines, we instead created questions based on them, as we didn't have complete descriptions of them at the time. The questions were as follows:

Learnabiltiy
Skipped due to overlapping with Exploration and Learnability.

Navigation
- Do the groupings of functions make sense? Do you think others think it makes sense?
- How does the programme allow you to navigate to a function if you don't know where it is?
- Are functions that are more often used easier to access than those that are used less often?

Language
- Read through the different menus. Do you understand the functions based on their names?
- Are there real representations for the different functions? Is there any that aren't?
- Does the programme's language fit in with the language of an image editing software?

Exploration and Learnability
- Do you feel anxious about exploring the programme? If no, What does the programme do to make you feel safe? If yes, Why does the programme make you feel anxious?
- Do you feel that you can learn from the resources that the programme provides? Do you think others would?
- Can you try to find a specific function? Do you feel as though the programme guided you? Do you understand why the function is where it is?

[1]**Found at: https://www.nngroup.com/articles/ten-usability-heuristics/**

Expectation of Image Editing Software

- What does the programme expect one should know about image editing? Why do you feel that?
- Do the icons and names resemble other photo editing software?
- Do you understand the different menus?

In total, we tested 2 users. In preparation, we found an image with the idea that the user could edit it as needed to learn better about the system. In our case, it turned out that the users needed a reason to interact with the image, something we recommend you do through scenarios. Even then, they gave a good amount of information relating to the interface itself, clearly explaining a lot of thoughts, both positive and negative, they had about the system.

Evaluating these tests gave us a lot of qualitative data, while it didn't give that much quantitative data. This led to the evaluation phase taking quite a bit longer than it should have, but through this method, we found ~8 issues per user, something far more than we found with a standard user test (where we found ~5 problems per user).

Outro

Interested in learning more?

This booklet is based on the thesis *A Lot with a Little - Evaluating Usability Methods for Developing Two New Methods for Small Test Groups*. If you are interested in the creation of these two methods, you can find both the theory and methods used in that document.

Thank you for reading, and we hope that you found this booklet both interesting and educational. If you have any questions, outside of reading the main document, you can contact either of the authors through LinkedIn:

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